Closing the Sim-to-Real Gap: Structure-Aware Dynamic Causal Calibration for Digital Twin Simulations

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Abstract

Maintaining fidelity between a stochastic simulator and the physical system it represents is a fundamental challenge for digital twins. Existing calibration approaches often correct simulator outputs independently, improving predictive accuracy while potentially violating structural relationships encoded by the simulator. We propose *Dynamic Causal Calibration* (DCC), a structure-aware framework that calibrates selected internal mechanisms through parametric interventions: it uses the simulator's dependency graph to identify parsimonious candidate intervention sets, learns horizon-dependent calibration functions that are shared across simulated entities and linked through a Bayesian state-space model, and then regenerates downstream quantities through the unchanged simulator equations. This design preserves the relationships encoded by those equations, while the number of calibration parameters scales with the number of calibrated mechanisms and prediction horizons and therefore does not grow with the number of simulated entities. We evaluate DCC in fully synthetic, semi-synthetic, and industrial experiments. In the synthetic benchmark, DCC reduces mean absolute error by 83–96% across reported quantities while preserving the tested structural identity. Independent output calibration attains comparable marginal accuracy on the non-recursive quantities but violates that identity for nearly every instance, and it is less accurate than DCC on the recursively propagated inventory level, where DCC's advantage widens with the prediction horizon. The semi-synthetic experiments show that corrections to selected upstream mechanisms propagate through a nonlinear simulator and improve downstream targets. In Amazon's inbound supply-chain digital twin, DCC reduces mean absolute error in purchase-order quantities by up to 37% and retains these improvements over five later evaluated system snapshots without retraining. Together, the experiments show that DCC can narrow target sim-to-real gaps while preserving the simulator relationships through which those targets are generated. They also reveal trade-offs in coupled non-target quantities when the retained simulator structure cannot reconcile all observed discrepancies simultaneously.

Keywords: dynamic causal calibration; simulation calibration; sim-to-real gap; structural equation model; discrete-event simulation; digital twin; Bayesian state-space model; supply chain

1 Introduction

Stochastic simulation is widely used to represent complex systems whose behavior emerges from interacting components, uncertain inputs, and dynamic state transitions. Increasingly, these simulators serve as *digital twins*: virtual representations that are repeatedly aligned with evolving physical systems and used for forecasting, planning, and what-if analysis. Their value depends not only on the accuracy of individual predictions but also on the coherence of the system state they produce. Simulated quantities are often linked by accounting identities, conservation relationships, and state-transition equations; forecasts that appear accurate in isolation may therefore be mutually inconsistent.

This problem complements established research on simulation fidelity. Work on input uncertainty distinguishes error due to finite simulation effort from uncertainty due to estimated input models and develops methods for quantifying the latter Barton et al. (2014); Nelson et al. (2021). Research on digital twins further emphasizes the need to maintain alignment between a virtual representation and an evolving physical system Rhodes-Leader and Nelson (2023). We study a different but related source of discrepancy: systematic error in the mechanisms encoded by the simulator. Such discrepancies can persist even when input distributions are carefully estimated. Motivated by Nelson's foundational work on simulation fidelity, we extend this research tradition to the calibration challenges posed by contemporary industrial digital twins. As his academic descendant, the second author is particularly honored to contribute to this Special Issue.

Mechanism discrepancy creates a tension between predictive accuracy and structural consistency. Correcting reported outputs independently may reduce their marginal errors while breaking the relationships through which they are jointly generated. Leaving the simulator unchanged preserves those relationships but retains systematic bias. This tension motivates our central question: *How can a structured stochastic simulator be calibrated to real-world observations while preserving the relationships encoded by the mechanisms that are not recalibrated?*

Existing approaches do not directly resolve this tension. Classical Bayesian calibration often treats a simulator as a monolithic input–output mapping, possibly through a surrogate model Kennedy and O'Hagan (2001). Output-calibration methods adjust predictions in a post-hoc manner after simulation Platt et al. (1999); Gneiting et al. (2005). Although effective for many predictive tasks, these approaches do not generally use the simulator's internal dependency structure to determine where a correction should enter or how it should propagate. This distinction becomes important when many reported quantities share upstream mechanisms and must be jointly modeled.

Large-scale supply-chain simulation provides a demanding example. Forecasts of purchase orders, inbound arrivals, and inventory positions support procurement, labor, and capacity decisions across multiple planning horizons. Errors can propagate from one stage to the next and accumulate over time. The corresponding quantities are also structurally coupled. For example, a supply-chain simulator may distinguish purchase orders (POs) already placed but not yet received (*Open PO*) from those expected to be placed in future planning periods (*Future PO*). Their sum is the total outstanding and expected purchase-order volume (*Total PO*), giving the identity

$$\text{Total PO} = \text{Open PO} + \text{Future PO},$$

along with an inventory transition such as

$$\text{Onhand}_t = \max(0, \text{Onhand}_{t-1} + \text{Inbound}_t - \text{Outbound}_t)\,.$$

Independent corrections can violate the first relationship and produce inventory trajectories incompatible with the second. A practical calibration method must also accommodate horizon-dependent bias while remaining scalable to large populations of simulated entities.

To address this problem, we adopt a causal formulation of the stochastic simulation system. Viewing a simulator as a causal object has precedent. Structural equation models and the do-calculus supply the intervention semantics (Pearl, 2009; Peters et al., 2017); continuous-time dynamical systems admit structural causal representations in which interventions modify the governing equations (Rubenstein et al., 2016; Bongers et al., 2018); and recent work argues that simulators implicitly define causal models of the systems they represent, so that their use for what-if analysis is a causal rather than a purely statistical problem (Dyer et al., 2024). The theory of causal abstraction further licenses reasoning at the granularity of simulator modules rather than individual internal state variables (Rubenstein et al., 2017; Beckers and Halpern, 2019), which is the level at which our dependency graph is elicited. We accordingly view the stochastic simulation system as a causal structural equation model: the system's component dependency gives rise to a directed causal graph, and each system module is a queryable black-box function with observed inputs, observed outputs, and added random noise. Under this framework, we propose **Dynamic Causal Calibration (DCC)**. DCC represents calibration as a parametric intervention at selected nodes of the simulator's directed dependency graph. Rather than correcting every reported output marginally for every instance, it learns universal calibration functions over an intervention subset and regenerates their counterfactual descendants following the dependency graph and the simulator equations. Horizon-specific parameters are modeled through a Bayesian state-space model, accounting for system discrepancies that evolve with forecast horizon, which in itself is coherent to the causal dependency. Here, *causal* refers to intervention semantics within the simulator: its graph records computational dependencies and is not assumed to be a fully identified causal graph of the physical system.

DCC is formulated for structured stochastic simulators with known computational dependencies; supply-chain simulation provides the motivating application and empirical testbed. The paper makes three contributions.

- We develop a structurally consistent formulation of simulation calibration. We view the stochastic simulation system as a structural equation model through a causal framework and formulate calibration as parametric intervention entered through parsimonious system components. To reduce the sim-to-real gap, elements of the selected intervention set are calibrated, while their causal descendants are counterfactually generated through the dependency graph and the original system equations. We further characterize, at the population level, conditions under which calibration through the selected mechanisms can remove the sim-to-real discrepancy and the estimation objective recovers the corresponding parameters (Section 3).

- We propose Dynamic Causal Calibration (DCC), a scalable calibration framework for large-scale dynamic simulation systems. Calibration functions are parameterized within a shared parametric

family and linked across prediction horizons through a Bayesian state-space model. This replaces the nonparametric entity-specific interventions of Aglietti et al. (2020); Singh (2023) with finitely many estimable quantities, which allows computation to scale with large sample sizes.

- We validate DCC at controlled and industrial scale. We evaluate DCC in fully synthetic and semi-synthetic experiments and in an industrial supply-chain digital twin covering 5,520 high-volume products over weekly prediction horizons. In the synthetic benchmark, DCC reduces MAE by 83–96% while preserving the tested structural identity, which independent output calibration violates for nearly every instance. In the industrial study, DCC reduces MAE by 27.6% for the targeted Total PO and by 36.6% for the calibrated Future PO, and retains target improvements over five later evaluated snapshots without retraining. The experiments also reveal trade-offs in coupled non-target outcomes, clarifying the limits of calibration through the retained simulator structure.

The remainder of this paper is organized as follows. Section 2 reviews related work on simulation calibration, causal modeling, and related simulation methodologies. Section 3 formulates the general DCC framework for structured stochastic simulators, and Section 4 presents its computational procedure. Section 5 instantiates DCC in the motivating supply-chain digital twin and describes the corresponding structural calibration problem. Section 6 evaluates DCC through fully synthetic, semi-synthetic, and industrial experiments. Finally, Section 7 summarizes the main findings and discusses limitations and directions for future research.

2 Related Work

2.1 Supply Chain Simulation and Digital Twins

Discrete-event simulation has a long history in supply chain management, from modeling individual warehouse operations to end-to-end network planning Terzi and Cavalieri (2004). Recent surveys document the growing adoption of DES for supply chain risk management, demand planning, and logistics optimization Maidana-Aquino et al. (2026). Digital twins extend this paradigm by maintaining a continuously updated virtual replica of the physical system, enabling what-if analysis and real-time decision support Taylor et al. (2023). However, the fidelity of these digital twins depends critically on calibration: even well-designed simulations exhibit systematic biases due to imperfect input distributions, simplified structural assumptions, and unmodeled external factors. The calibration challenge is especially acute in supply chain settings where (1) the system evolves continuously (new vendors, changing logistics networks, shifting demand patterns), limiting the usefulness of historical data for purely statistical corrections, and (2) multiple output metrics are structurally coupled through physical identities that must be preserved.

2.2 Simulator Calibration

The foundational framework of Kennedy and O'Hagan (2001) models the discrepancy between simulator and reality as an additive Gaussian process (GP), estimating unknown calibration parameters via Bayesian inference. Extensions include multivariate output calibration Higdon et al. (2008), online fixed-inducing-point

methods for streaming data Duan et al. (2021), and deep Gaussian processes for improved scalability Marmin and Filippone (2022). These approaches treat the simulator as a monolithic input-output mapping and construct a surrogate emulator, typically a GP, to approximate the black-box function. While principled, they do not exploit the internal causal structure of the simulation, nor do they scale to large populations of heterogeneous instances that each require consistent multi-metric forecasts.

Data assimilation methods, originally developed for weather forecasting and physical systems, have recently been adapted for online calibration of DES models. Hu and Yan (2024) present a particle filter-based approach for assimilating real-time observations into simulation digital twins, enabling sequential parameter updates as new data arrive. While effective for online tracking of slowly drifting parameters, data assimilation methods generally do not exploit the DAG structure of the simulator and do not enforce structural identities across coupled outputs. Recent work on online Bayesian calibration under non-stationarity addresses gradual drift and abrupt regime shifts by separating a sequential particle update for calibration parameters from a Gaussian process update for discrepancy, augmented with restart mechanisms that detect regime shifts (Xu and Park, 2026), but likewise operates without causal structural constraints.

The emerging digital twin literature recognizes calibration as a first-class concern. Zheng et al. (2025) propose the Actor-Simulator framework, which jointly calibrates a digital twin and searches for optimal control policies via model-based reinforcement learning. Jeon et al. (2025) survey stochastic simulation calibration in the digital twin era, identifying the need for methods that handle continuous data ingestion and non-stationary environments. Our work complements these efforts by providing a calibration layer that preserves the simulator's causal structure, which is orthogonal to how the calibrated simulator is subsequently used for decision-making.

A parallel literature addresses the uncertainty introduced by estimating simulation input distributions from finite real-world data. Barton et al. (2014) provide confidence intervals that account for input uncertainty, while Nelson et al. (2021) propose model averaging to reduce input-model risk. These methods focus on quantifying and reducing the uncertainty in *input distributions* rather than correcting systematic discrepancies in *output metrics*. Our framework addresses a complementary problem: given a simulator with fixed input models, how to correct the resulting output bias while preserving structural consistency.

2.3 Metamodel-Based Simulation Optimization

Metamodels, or surrogate models, approximate the input–output behavior of expensive simulations to enable efficient optimization. Barton and Meckesheimer (2006) survey polynomial, kriging, and neural network metamodels for simulation optimization, and Barton (2009) provides a tutorial on using metamodels for discrete-event simulation optimization. Stochastic kriging extends Gaussian process metamodels to handle the intrinsic noise of stochastic simulations (Ankenman et al., 2010). These methods share with our work the goal of making simulation-based decisions computationally tractable, but they focus on optimizing *decision variables*, for example inventory policies or staffing levels, rather than calibrating *structural mechanisms* to reduce the sim-to-real gap.

Three features of a large industrial simulator also limit the direct application of the monolithic view these methods adopt, in which the simulator is replaced by a single response surface. First, the argument of that

response surface is the full system state across the product catalog, so the mapping to be approximated is high-dimensional and heterogeneous rather than a smooth surface over a handful of design variables. Second, the object of interest is not one performance measure but a collection of coupled quantities reported for every entity and every prediction horizon, which an end-to-end surrogate would have to reproduce jointly and consistently. Third, the modules of a production simulator are typically owned and maintained by different engineering teams and evolve on independent schedules, so a surrogate that replaces the simulator wholesale is difficult to validate against its components and harder still to keep current as any one of them changes.

We therefore emulate at the level of modules rather than end to end. The emulator used in our experiments replaces selected computationally expensive mechanisms with learned approximations while retaining the surrounding discrete-event logic and the dependency structure among modules (Lu et al., 2025), and Section 5.1 describes it in detail. This choice is what makes the structural view of Section 3 available at all. A simulator abstracted module by module still exposes the intermediate quantities through which a correction propagates, so calibration can enter upstream and be carried forward by the retained equations; a single input–output surrogate exposes only its endpoints, leaving nothing to intervene on but the reported outputs. In this sense module-level emulation and structure-aware calibration are complementary choices about the same question, namely how much of the simulator's internal structure to keep.

DCC could in principle be combined with metamodel-based optimization in a two-stage pipeline: first calibrate the simulator via DCC, then optimize decisions using the calibrated simulator.

2.4 Causal Modeling and Intervention

Structural equation models (SEMs) and the do-calculus of Pearl (2009) provide a formal language for reasoning about interventions. Peters et al. (2017) extend this to general causal inference. Bongers et al. (2018) and Rubenstein et al. (2016) formalize how continuous-time dynamical systems (ODEs) can be cast as SCMs, where interventions modify the governing equations. Our work applies this causal perspective to discrete-event simulators, where the "structural equations" are the simulation's modular logic components (demand models, procurement logic, inventory transitions).

Closest to our optimization formulation is Aglietti et al. (2020), which introduces Causal Bayesian Optimization (CBO) for optimizing *hard* interventions on a known DAG to improve a target outcome. CBO selects which variables to intervene on and optimizes instance-specific intervention values. Singh (2023) extends this to nonparametric settings. Our parametric formulation differs in two key respects: (1) we use *parametric (soft) interventions* that learn shared calibration functions across all instances, rather than instance-specific hard interventions, reducing the parameter count from $O(N \cdot |\mathscr{C}|)$ to $O(H \cdot |\mathscr{C}|)$; and (2) we optimize an inverse-intervention objective that minimizes sim-to-real gaps across multiple coupled outcomes simultaneously, rather than optimizing a single target.

Finally, Kocaoglu et al. (2017) study cost-optimal selection of intervention sets for causal graph identification. We repurpose their antichain enumeration algorithm for a different goal: constructing parsimonious candidate calibration sets that avoid redundant interventions along the same directed path (Section 4.1).

2.5 Output and Probability Calibration

Calibration of predictive outputs has a rich history in machine learning. Platt scaling calibrates probability estimates via logistic regression Platt et al. (1999), with online extensions Gupta and Ramdas (2023). For non-negative count data, log-space linear rescaling provides interpretable mean corrections Gneiting et al. (2005); Green (2021). Isotonic distributional regression offers nonparametric calibration of full conditional distributions Henzi et al. (2021). Our calibration function (Section 5.3) can be viewed as a log-space linear map in this tradition, but embedded within a causal DAG and jointly optimized across multiple structurally coupled metrics rather than applied independently to each output.

3 Dynamic Causal Calibration: General Formulation

Dynamic Causal Calibration (DCC) treats simulation calibration as a *structural intervention problem*. Rather than correcting reported outputs after simulation, DCC modifies selected mechanisms within the simulator and propagates their effects through the simulator's existing dependency structure. This distinction is central: the objective is not only to reduce discrepancy between simulated and observed quantities, but to do so without discarding the structural relationships that make the simulator a coherent representation of the underlying system.

The framework consists of three components. First, we represent the simulator as a structural equation model (SEM), making its computational dependencies explicit. Second, we identify a parsimonious set of mechanisms at which calibration can enter the simulator. Third, we learn shared parametric interventions that minimize sim-to-real discrepancy while allowing corrections to propagate through the remaining structural equations. This section develops the first component together with the calibration model and objective; Section 4 presents the computational procedure for selecting where to calibrate and estimating the calibration parameters. Figure 1 summarizes the DCC workflow, which consists of three stages: locating candidate mechanisms, calibrating selected mechanisms, and propagating the resulting corrections through the retained simulator structure.

3.1 Problem Setting and Assumptions

DCC distinguishes the physical system, the original simulator, and the calibrated simulator. At system snapshot t, the original simulator generates N entity-level realizations over the prediction horizons $\mathscr{H} = \{0, \ldots, H-1\}$. Real-world observations are available for a subset of quantities, and only a subset of simulator components can feasibly be modified. DCC seeks a calibrated simulator that improves agreement with the observed quantities while leaving every non-intervened simulator equation unchanged.

The framework relies on four assumptions.

A1. The simulator's computational dependencies are known and can be represented by a directed acyclic graph after unrolling any temporal recursion over the prediction horizon.

A2. Candidate interventions can be inserted before affected descendants are evaluated.

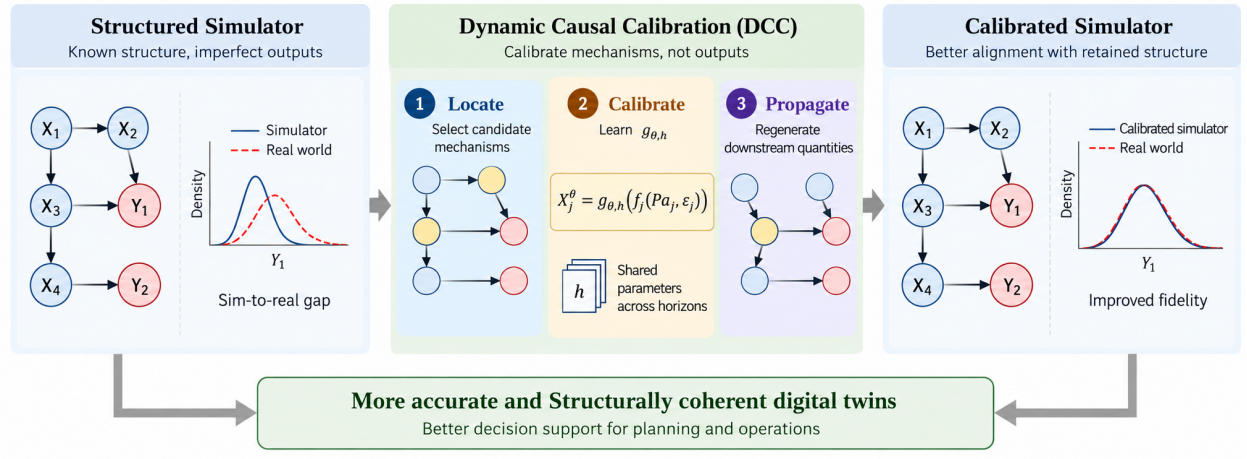


Figure 1: **Overview of Dynamic Causal Calibration (DCC).** DCC identifies candidate mechanisms associated with sim-to-real discrepancy, learns horizon-dependent calibration functions, and propagates the resulting corrections through the retained simulator structure. The resulting calibrated simulator improves fidelity while preserving structural relationships encoded by non-intervened mechanisms.

A3. The observations used to fit and evaluate calibration can be aligned with the corresponding simulator realizations.

A4. The chosen parametric calibration family is shared across entities, although its parameters may vary with prediction horizon.

The simulator may otherwise be stochastic, non-differentiable, and accessible only through function evaluations.

3.2 Simulation as a Structural Equation Model

Table 1 summarizes the notation used throughout the methodology. We write V for a generic variable of the SEM, X for a member of the calibration set, Y for an outcome, and Z for an observed quantity of either kind.

Definition 1 (Structural Equation Model)**.** A structural equation model is a tuple $\mathscr{M} = (\mathscr{U}, \mathscr{V}, \mathscr{F}, P_{\mathscr{U}})$, where $\mathscr{U}$ is a set of exogenous variables with joint distribution $P_{\mathscr{U}}$, $\mathscr{V}$ is a set of endogenous variables, and $\mathscr{F} = \{f_V : V \in \mathscr{V}\}$ is a collection of structural equations

$$V := f_V(\mathrm{Pa}(V), U_V),$$

with $\mathrm{Pa}(V) \subseteq \mathscr{V} \setminus \{V\}$ denoting the endogenous parents of V and $U_V \subseteq \mathscr{U}$ its exogenous inputs (Pearl, 2009). The SEM induces a directed acyclic graph $\mathscr{G}$ over $\mathscr{V}$, with edge $V \to W$ whenever $V \in \mathrm{Pa}(W)$.

A stochastic simulator naturally admits this representation when its modular computational logic is known. Exogenous inputs and simulation randomness form $\mathscr{U}$; simulated states and outputs form $\mathscr{V}$; and the simulator's transition, accounting, and decision rules form $\mathscr{F}$. Thus, the DAG should be interpreted primarily as the simulator's *structural or computational dependency graph*: it records how quantities are generated

Table 1: Notation for the DCC framework.

Symbol	Type	Meaning
h	Index	Prediction-horizon index, $h \in \mathscr{H} = \{0, \dots, H-1\}$.
t	Index	Time index of the system snapshot.
i	Index	Index of simulated entities, $i = 1, \dots, N$.
$\mathscr{M}$	SEM	Structural equation model $\mathscr{M} = (\mathscr{U}, \mathscr{V}, \mathscr{F}, P_{\mathscr{U}})$.
$\mathscr{G}$	DAG	Directed acyclic graph induced by $\mathscr{F}$.
$\mathscr{U}$	Set	Exogenous variables determined outside the model.
$\mathscr{V}$	Set	Endogenous variables determined by $\mathscr{F}$.
$\mathscr{F}$	Set	Structural equations mapping parents and exogenous inputs to endogenous variables.
$\mathrm{Pa}(V)$	Set	Endogenous parents of V; $\mathrm{Pa}^{\boldsymbol{\theta}}(V)$ denotes their values after upstream calibration.
$\mathscr{O}$	Set	Outcomes with observed sim-to-real discrepancy, $\mathscr{O} \subseteq \mathscr{V}$.
$\mathscr{I}$	Set	Candidate variables at which calibration may intervene, $\mathscr{I} \subseteq \mathscr{U} \cup \mathscr{V}$.
$\mathscr{C}$	Set	Selected calibration set, $\mathscr{C} \subseteq \mathscr{I}$.
$\mathscr{O}^{in}$	Set	Calibrated variables with observed real-world counterparts (inputs of the calibration), $\mathscr{O}^{in} \subseteq \mathscr{C}$.
f_V	Function	Structural equation associated with endogenous variable V.
$g_{\boldsymbol{\theta}_V}$	Function	Parametric calibration function applied to $V \in \mathscr{C}$.
$\boldsymbol{\theta}$	Parameters	Calibration parameters, $\boldsymbol{\theta} = \{\boldsymbol{\theta}_V : V \in \mathscr{C}\}$, with horizon-specific components $\boldsymbol{\theta}_{V,h}$.
$x^{\boldsymbol{\theta}}_{i,h}, x^*_{i,h}$	Values	Calibrated and observed values of $X \in \mathscr{O}^{in}$ for entity i at horizon h.
$y^{\boldsymbol{\theta}}_{i,h}, y^*_{i,h}$	Values	Calibrated and observed values of outcome $Y \in \mathscr{O}$ for entity i at horizon h.

within the simulator and therefore how a change introduced at one component propagates to others. It is not, without additional assumptions, an identified causal graph of the physical system.

3.3 Calibration as Parametric Intervention

Definition 2 (Parametric Intervention)**.** A *soft intervention* on V replaces its structural mechanism (Peters et al., 2017),

$$V := f_V(\mathrm{Pa}(V), U_V) \qquad \longrightarrow \qquad V := f'_V(\mathrm{Pa}(V), U_V).$$

When the parent set is preserved, the modification is a *parametric intervention* (Eberhardt and Scheines, 2007; Eaton and Murphy, 2007). The hard intervention $\mathrm{do}(V = a)$ is recovered as the special case in which f'_V is the constant function a (Pearl, 2009).

In the implementation considered here, calibration transforms the value produced by a selected simulator component before any descendants are evaluated. Thus, for $V \in \mathscr{C}$,

$$f'_V(\mathrm{Pa}(V), U_V) = g_{\boldsymbol{\theta}_V}(f_V(\mathrm{Pa}(V), U_V)).$$

This is an internal mechanism replacement rather than post-hoc correction of a reported descendant: the transformed value becomes part of the simulator state and is consumed by all downstream equations.

DCC uses this intervention language in an inverse direction. Classical causal optimization asks how an

intervention changes an outcome. Here, observations of the physical system reveal a discrepancy in one or more outcomes, and the calibration problem asks which simulator mechanisms should be modified, and by how much, so that the resulting simulated system better agrees with those observations.

At snapshot time t and prediction horizon $h \in \mathscr{H}$, suppose that the simulator produces N realizations. We distinguish three sets. The outcome set $\mathscr{O} \subseteq \mathscr{V}$ contains quantities for which real-world observations are available and calibration performance is evaluated. The candidate intervention set $\mathscr{I} \subseteq \mathscr{U} \cup \mathscr{V}$ contains mechanisms that can feasibly be modified. Candidate interventions may also enter at exogenous variables: an intervention on $U \in \mathscr{U}$ replaces its marginal distribution, which is coherent with the SEM formulation. DCC selects a calibration set $\mathscr{C} \subseteq \mathscr{I}$, and $\mathscr{O}^{in} \subseteq \mathscr{C}$ denotes calibrated quantities that additionally have direct real-world observations and can therefore serve as calibration anchors; the superscript "in" indicates that these variables are inputs of the calibration, in contrast to the outcomes in $\mathscr{O}$.

For $X \in \mathscr{C}$, let $X^{\boldsymbol{\theta}} = g_{\boldsymbol{\theta}_X}(X)$ denote its calibrated value. All non-intervened descendants are then recomputed using their original structural equations.

Remark 1 (Calibration within the simulator). The distinction between DCC and output-wise correction is therefore architectural rather than merely statistical. DCC changes selected mechanisms *inside* the simulator and then allows the simulator to regenerate downstream quantities. Consequently, structural equations that are not intervened upon remain part of the calibrated system. This is the source of DCC's structural-consistency advantage: consistency is inherited from the simulator rather than imposed afterward through an additional reconciliation step.

Proposition 1 (Preservation of retained simulator equations)**.** Fix an intervention set $\mathscr{C}$. If every node $V \notin \mathscr{C}$ is evaluated with its original structural function f_V using the calibrated values of its parents, then every deterministic equation associated with a non-intervened node is satisfied by every realization of the calibrated simulator.

Proof. Evaluate the unrolled graph in topological order. For each $V \in \mathscr{C}$, DCC computes the intervened value using $g_{\boldsymbol{\theta}_V} \circ f_V$. For each $V \notin \mathscr{C}$, all parent values have already been computed and DCC assigns $V^{\boldsymbol{\theta}} = f_V(\mathrm{Pa}^{\boldsymbol{\theta}}(V), U_V)$. Hence the original equation for V holds by construction. Repeating this argument in topological order establishes the result for every non-intervened node. □

Proposition 1 is deliberately limited. DCC preserves equations retained by the calibrated simulator; it does not preserve the original equation at an intervened node, guarantee relationships absent from the simulator, or establish that the retained equations correctly describe the physical system.

Toy example. Consider the simulator depicted in Figure 1, with internal variables $X_1, \ldots, X_4$, outcomes Y_1, Y_2, and structural equations

$$X_1 := U_{X_1}, \quad X_2 := f_{X_2}(X_1, U_{X_2}), \quad X_3 := f_{X_3}(X_1, U_{X_3}), \quad X_4 := f_{X_4}(X_3, U_{X_4}),$$

$$Y_1 := X_2 + X_3, \qquad Y_2 := f_{Y_2}(X_4, U_{Y_2}).$$

Suppose observations reveal systematic discrepancy in both outcomes. Output calibration corrects each reported quantity with its own map, and the corrected values need not satisfy $Y_1 = X_2 + X_3$. DCC instead selects the calibration set $\mathscr{C} = \{X_2, X_3\}$ and replaces the two mechanisms by $g_{\boldsymbol{\theta}_{X_2}} \circ f_{X_2}$ and $g_{\boldsymbol{\theta}_{X_3}} \circ f_{X_3}$. The simulator regenerates X_4, Y_1, and Y_2 through their original equations: the identity $Y_1 = X_2 + X_3$ holds for every realization by Proposition 1, and the correction at X_3 propagates to both outcomes. Two learned functions serve all realizations. The purchase-order identity in Section 1 has the same additive structure.

3.4 Calibrated Model and Objective

For each $V \in \mathscr{C}$, DCC replaces its mechanism by

$$V^{\boldsymbol{\theta}} = g_{\boldsymbol{\theta}_V}(V), \qquad \boldsymbol{\theta} = \{\boldsymbol{\theta}_V : V \in \mathscr{C}\}.$$

Every non-intervened node is evaluated in topological order using its original structural function,

$$V^{\boldsymbol{\theta}} = f_V\left(\mathrm{Pa}^{\boldsymbol{\theta}}(V), U_V\right), \qquad V \notin \mathscr{C},$$

where $\mathrm{Pa}^{\boldsymbol{\theta}}(V)$ denotes parent values after any upstream calibration has propagated to them.

When the exogenous inputs are jointly independent across mechanisms and the induced distributions admit conditional densities (or, more generally, regular conditional probability kernels), the joint distribution of the calibrated simulator factorizes as

$$p_{\boldsymbol{\theta}}(\mathbf{v}) = \prod_{V \in \mathscr{C}} p_{\boldsymbol{\theta}}\left(v_V \mid v_{\mathrm{Pa}(V)}\right) \prod_{V \notin \mathscr{C}} p\left(v_V \mid v_{\mathrm{Pa}(V)}\right). \tag{1}$$

The first product contains the mechanisms replaced by calibration, whereas the second contains the original simulator mechanisms. Equation (1) is a structural characterization of the calibrated simulator; it is not a requirement that every internal factor be available as a tractable likelihood. For deterministic or black-box components, the corresponding relationships are enforced by executing the simulator.

Inverse intervention objective

For a fixed candidate set $\mathscr{C}$, DCC chooses parameters that make observed outcomes and calibration anchors agree with the physical system. Let $y_{i,h}^{\boldsymbol{\theta}}$ and $x_{i,h}^{\boldsymbol{\theta}}$ denote quantities generated by the calibrated simulator for entity i and horizon h, and let $y_{i,h}^{*}$ and $x_{i,h}^{*}$ denote their observed counterparts. Let $\ell_Z(z^*, z^{\boldsymbol{\theta}})$ be the negative log-likelihood, or another prespecified loss, for observed quantity Z. The finite-sample MAP objective used by DCC has the form

$$\widehat{\boldsymbol{\theta}}_{\mathscr{C}} = \arg\min_{\boldsymbol{\theta}} \left\{ \sum_{h \in \mathscr{H}} \sum_{i=1}^{N} \left[\sum_{Y \in \mathscr{O}} \ell_Y\left(y_{i,h}^{*}, y_{i,h}^{\boldsymbol{\theta}}\right) + \sum_{X \in \mathscr{O}^{in}} \ell_X\left(x_{i,h}^{*}, x_{i,h}^{\boldsymbol{\theta}}\right) \right] - \log p(\boldsymbol{\theta}) \right\}. \tag{2}$$

Here $p(\boldsymbol{\theta})$ is the parameter prior. Application-specific likelihoods and parameter constraints must be specified before fitting. Section 5 provides these choices for non-negative supply-chain volumes.

Equation (2) does not include likelihood factors for every internal simulator node. Deterministic and stochastic simulator equations are enforced by executing the calibrated simulator: intervened nodes use their modified mechanisms, and all other nodes use their original mechanisms. This separation makes the statistical objective operational even when the simulator contains deterministic equations or does not expose tractable internal densities.

3.5 Theoretical Properties

Proposition 1 holds for every parameter value: retained equations are never violated. It does not address whether some parameter value removes the sim-to-real discrepancy, nor whether the estimation objective finds such a value. The following result answers both questions at the population level. Recall from Assumption A1 that temporal recursion is unrolled, so each variable in $\mathscr{G}$ has one copy per horizon $h \in \mathscr{H}$; for $X \in \mathscr{C}$, calibration applies $g_{\boldsymbol{\theta}_{X,h}}$ to the horizon-h copy of X.

The first two conditions relate the simulator $\mathscr{M} = (\mathscr{U}, \mathscr{V}, \mathscr{F}, P_{\mathscr{U}})$ to the physical system.

C1. The physical system admits an SEM representation $\mathscr{M}^* = (\mathscr{U}, \mathscr{V}, \mathscr{F}^*, P_{\mathscr{U}})$ inducing the same unrolled graph $\mathscr{G}$, with $f_V^* = f_V$ for every $V \notin \mathscr{C}$.

C2. For each $X \in \mathscr{C}$ and each $h \in \mathscr{H}$, there exists a parameter $\boldsymbol{\theta}^*_{X,h}$ of the calibration family such that the horizon-h mechanism of X satisfies $f_X^* = g_{\boldsymbol{\theta}^*_{X,h}} \circ f_X$.

Condition C1 states that the sim-to-real discrepancy originates only in the mechanisms of $\mathscr{C}$; condition C2 states that the calibration family is rich enough to express that discrepancy at every horizon. The remaining two conditions concern the estimation objective.

C3. Distinct parameter values induce distinct collections of laws of the observed quantities: if $\boldsymbol{\theta} \neq \boldsymbol{\theta}'$, then for some $Z \in \mathscr{O} \cup \mathscr{O}^{in}$ and $h \in \mathscr{H}$ the laws of $z_h^{\boldsymbol{\theta}}$ and $z_h^{\boldsymbol{\theta}'}$ differ.

C4. For each $Z \in \mathscr{O} \cup \mathscr{O}^{in}$ and $h \in \mathscr{H}$, the population loss $\mathbb{E}\big[\ell_Z(z_h^*, z_h^{\boldsymbol{\theta}})\big]$, with expectation over entities and system randomness, is minimized over $\boldsymbol{\theta}$ exactly at those values for which the law of $z_h^{\boldsymbol{\theta}}$ coincides with the law of z_h^*.

Condition C3 is an identification assumption; it holds or fails depending on the calibration set, the observed quantities, and the calibration family. Condition C4 holds, for example, when ℓ_Z is the negative log-likelihood of a correctly specified observation model. The population version of the objective is the per-entity expectation of its loss terms: as the number of entities grows, the loss terms in Equation (2) scale with N while the prior term $-\log p(\boldsymbol{\theta})$ remains fixed. The prior therefore acts as finite-sample regularization and does not appear in the population limit.

Theorem 1 (Calibratability and population recovery)**.** Suppose C1 and C2 hold, and set $\boldsymbol{\theta}^* = \{\boldsymbol{\theta}^*_{X,h} : X \in \mathscr{C}, h \in \mathscr{H}\}$.

(i) The calibrated simulator with parameters $\boldsymbol{\theta}^*$ induces the same joint distribution over $\mathscr{V}$ as the physical system $\mathscr{M}^*$. In particular, the joint distribution of the observed quantities in $\mathscr{O} \cup \mathscr{O}^{in}$ coincides with that of the physical system.

(ii) If, in addition, C3 and C4 hold, then $\boldsymbol{\theta}^*$ is the unique minimizer of the population version $R(\boldsymbol{\theta})$ of the objective in Equation (2) with the prior term omitted, and the calibrated simulator at the minimizer coincides with $\mathscr{M}^*$: its mechanisms equal $\mathscr{F}^*$ and it induces the same joint distribution over $\mathscr{V}$.

Proof. Part (i). By C1, the two systems share the exogenous variables $\mathscr{U}$ and their distribution $P_{\mathscr{U}}$. Couple them on a common probability space by evaluating both on the same realization of $\mathscr{U}$, and proceed by induction over a topological order of the unrolled graph $\mathscr{G}$. For a node $V \notin \mathscr{C}$, C1 gives $f_V^* = f_V$; by the induction hypothesis its parent values agree in the two systems, so its values agree. For the horizon-h copy of $X \in \mathscr{C}$, the induction hypothesis and C2 give

$$g_{\boldsymbol{\theta}^*_{X,h}}\Big(f_X(\mathrm{Pa}^{\boldsymbol{\theta}^*}(X), U_X)\Big) = f_X^*(\mathrm{Pa}^*(X), U_X),$$

so its values agree as well. Hence every node of the two systems agrees almost surely under the coupling, and the induced joint distributions over $\mathscr{V}$ coincide.

Part (ii). By part (i), at $\boldsymbol{\theta}^*$ the law of every observed quantity coincides with that of its real counterpart, so by C4 every term of R attains its minimum at $\boldsymbol{\theta}^*$. Hence $\boldsymbol{\theta}^*$ minimizes R, and $\min R$ equals the sum of the per-term minima. Let $\boldsymbol{\theta}'$ be any minimizer of R. Since every term of R is bounded below by its own minimum and the terms attain these minima simultaneously at $\boldsymbol{\theta}^*$, each term must attain its minimum at $\boldsymbol{\theta}'$ as well. By C4, the law of every observed quantity under $\boldsymbol{\theta}'$ coincides with the real law, and C3 gives $\boldsymbol{\theta}' = \boldsymbol{\theta}^*$. The mechanisms outside $\mathscr{C}$ equal $\mathscr{F}^*$ by C1, and at each $X \in \mathscr{C}$ and $h \in \mathscr{H}$, C2 gives $g_{\boldsymbol{\theta}^*_{X,h}} \circ f_X = f_X^*$. The distributional claim is part (i). □

Without C3 and C4, part (i) asserts existence only: several parameter values, or several calibration sets, may induce the same observed distributions, and the identifiability discussion in Section 4 applies unchanged.

Remark 2 (When no calibration reconciles the observations). Contrapositively, if no parameter value makes the joint distribution of the observed quantities match the physical system, then C1 or C2 fails for the chosen calibration set: the discrepancy cannot be expressed through the selected mechanisms and calibration family. A different calibration set may still satisfy C1–C2. This is the population counterpart of the trade-offs observed in the industrial experiments (Section 6.3), where discrepancies in coupled outcomes could not be reconciled simultaneously through the retained structure.

4 Computational Formulation of DCC

Section 3 formulated calibration as parametric intervention on the simulator's structural equations and defined the estimation objective. This section presents the computational procedure: constructing and comparing candidate calibration sets, parameterizing the interventions so that estimation scales to large simulations, linking calibration parameters across prediction horizons, and solving the resulting optimization problem.

4.1 Selecting Where to Calibrate

A discrepancy observed at an output need not originate at that output. Because errors propagate through the simulator, the same downstream discrepancy may be generated by several upstream mechanisms. Calibration therefore involves two coupled decisions: *where* to intervene and *how strongly* to intervene.

Searching over every subset of $\mathscr{I}$ is combinatorial and can also produce redundant calibration mechanisms. We therefore construct and compare candidate calibration sets using the following procedure:

1. **Root-cause screening.** Rank variables in $\mathscr{I}$ by their potential contribution to the observed sim-to-real discrepancy. The ranking may be obtained from domain knowledge, diagnostic models, or attribution methods such as Shapley values (Ng et al., 2025). The screening rule and the number k retained are fixed using training data and application constraints.

2. **Antichain enumeration.** Among the retained variables, enumerate nonempty antichains of $\mathscr{G}$ (Kocaoglu et al., 2017). An antichain contains no ancestor–descendant pair, thereby discouraging simultaneous interventions at multiple locations along the same directed path. Singleton sets are retained, and the empty set represents the uncalibrated simulator.

3. **Feasibility and observability filtering.** Remove candidate sets containing mechanisms that cannot be modified in the intended deployment setting. When a component is required to serve as a calibration anchor, it must also have an aligned real-world observation; unobserved components may remain candidates only when their effects can be learned through observed descendants.

4. **Validation-based selection.** Fit the calibration parameters for every surviving set on the estimation data and rank the sets on held-out validation data using a prespecified criterion, such as negative log-likelihood with an error metric used as a secondary criterion.

Antichain enumeration is therefore a structural pre-filter, not a claim that graph topology uniquely identifies the correct calibration set. Its role is to limit redundant explanations before statistical comparison on held-out data.

Remark 3 (Calibration location is part of the model.). The intervention set is not merely a computational tuning choice. It expresses a hypothesis about *where the simulator differs from the physical system*. Two calibration models can therefore fit the same observed outcome while implying different mechanisms of discrepancy and different downstream responses. DCC makes this modeling choice explicit rather than hiding it inside an unrestricted output correction.

4.2 Shared Parametric Calibration Functions

Once a calibration set $\mathscr{C}$ has been selected, DCC learns the magnitude and form of the interventions. Standard causal optimization (Aglietti et al., 2020; Singh, 2023) typically optimizes intervention values for individual instances. Such an approach becomes impractical when a simulator contains a large population of heterogeneous entities or is repeatedly evaluated over multiple horizons.

DCC instead learns a shared parametric calibration function $g_{\boldsymbol{\theta}_X}$ for each $X \in \mathscr{C}$. With a fixed-dimensional parameterization at each horizon, the number of calibration parameters scales as $O(H|\mathscr{C}|)$ rather than $O(N|\mathscr{C}|)$ under instance-specific interventions. The learned intervention therefore represents a reusable correction mechanism, not a separate adjustment for every simulation realization.

Remark 4 (Shared interventions learn mechanisms, not instances). The reduction from $O(N|\mathscr{C}|)$ to $O(H|\mathscr{C}|)$ is more than a computational device. Instance-specific corrections can memorize discrepancies associated with individual simulation realizations. A shared intervention instead asks whether a common transformation

of a simulator mechanism explains the sim-to-real discrepancy across the population. This distinction is important for deployment, because the learned calibration can be applied to new simulation instances without estimating new instance-level parameters.

4.3 Dynamic Calibration across Prediction Horizons

The sim-to-real discrepancy may change with forecast horizon. DCC therefore allows each calibrated mechanism to have horizon-specific parameters while partially pooling adjacent horizons. In general vector form, after transforming any constrained components of $\boldsymbol{\theta}_X$ to an unconstrained scale, the dynamic prior is

$$\boldsymbol{\theta}_{X,0} \sim \mathcal{N}(\boldsymbol{\theta}_X^0, Q_{X,0}), \tag{3}$$

$$\boldsymbol{\theta}_{X,h} \mid \boldsymbol{\theta}_{X,h-1} \sim \mathcal{N}(\boldsymbol{\theta}_{X,h-1}, Q_X), \qquad h = 1, \ldots, H-1, \tag{4}$$

for each $X \in \mathscr{C}$. The center $\boldsymbol{\theta}_X^0$ represents the identity transformation. The innovation covariance Q_X controls how rapidly the calibration may change across horizons: $Q_X = 0$ yields a common static correction, whereas weak pooling approaches separate horizon-by-horizon fits. This evolution is across look-ahead horizon within a system snapshot, not across calendar snapshot time. Section 5 specializes the prior to the log-linear calibration used in the supply-chain application.

4.4 End-to-End Procedure

Algorithm 1 summarizes the complete estimation and selection procedure. The validation criterion and all screening thresholds are fixed before the candidate sets are compared.

Algorithm 1 Dynamic Causal Calibration

Input: simulator $\mathscr{M}$ and its computational graph $\mathscr{G}$; prediction horizons $\mathscr{H}$; observed outcomes $\mathscr{O}$; feasible interventions $\mathscr{I}$; observed anchors $\mathscr{O}^{in}$; estimation data $\mathscr{D}_{\text{est}}$; validation data $\mathscr{D}_{\text{val}}$.

Output: selected calibration set $\widehat{\mathscr{C}}$ and fitted parameters $\widehat{\boldsymbol{\theta}}_{\widehat{\mathscr{C}}}$.

1: Unroll temporal recursions in $\mathscr{G}$ over $\mathscr{H}$ and align observed quantities with the corresponding simulator realizations.
2: Screen variables in $\mathscr{I}$ using the prespecified root-cause ranking and retain the top k feasible candidates.
3: Form the candidate collection $\mathfrak{C}$ from the feasible nonempty antichains of the screened graph; include $\varnothing$ as the uncalibrated baseline.
4: **for each** $\mathscr{C} \in \mathfrak{C}$ **do**
5: Execute the calibrated simulator on $\mathscr{D}_{\text{est}}$ and compute $\widehat{\boldsymbol{\theta}}_{\mathscr{C}}$ by solving Equation (2).
6: Evaluate $(\mathscr{C}, \widehat{\boldsymbol{\theta}}_{\mathscr{C}})$ on $\mathscr{D}_{\text{val}}$ using the prespecified validation criterion.
7: **end for**
8: Select the candidate with the best validation performance, $\widehat{\mathscr{C}} = \arg\min_{\mathscr{C} \in \mathfrak{C}} \mathscr{L}_{\text{val}}(\mathscr{C}, \widehat{\boldsymbol{\theta}}_{\mathscr{C}})$.
9: **return** $(\widehat{\mathscr{C}}, \widehat{\boldsymbol{\theta}}_{\widehat{\mathscr{C}}})$.

In practice, the simulator may be non-differentiable or accessible only through simulation evaluations. DCC therefore does not require gradients of the underlying simulator. We use COBYLA (Powell, 1994) as the default derivative-free optimizer and use CMA-ES (Hansen et al., 2019) as a more computationally intensive alternative when greater robustness to local search behavior is desired.

4.5 Interpretation and Identifiability

DCC identifies a correction that is compatible with the chosen simulator structure, parametric family, and available observations. These ingredients do not generally identify the unique physical cause of a discrepancy. Several intervention sets may produce similar observed outcomes, particularly when few internal quantities are observed. Predictive identification of calibrated outputs should therefore be distinguished from identification of the calibration parameters, the intervention location, or a causal mechanism in the physical system. Held-out comparison can select among useful predictive models, but a mechanistic interpretation requires additional observability or domain assumptions.

The framework above is independent of any particular application domain. Implementation requires specifying three application-dependent ingredients: the manipulable mechanisms $\mathscr{I}$, a parametric family $g_{\boldsymbol{\theta}}$ appropriate for the quantities being calibrated, and a likelihood model appropriate for the corresponding observations. In the next section, we instantiate these components for a large-scale supply-chain digital twin and then evaluate the resulting calibration in controlled and industrial experiments.

5 Supply-Chain Digital Twin and DCC Instantiation

The DCC framework in Sections 3 and 4 is defined for a general structured stochastic simulator. We now instantiate the framework in a large-scale supply-chain digital twin. This setting is particularly useful for studying structural calibration because the simulator contains both complex stochastic mechanisms and deterministic relationships that must remain internally consistent. It also provides repeated system snapshots, allowing us to evaluate whether a calibration learned at one point in time remains useful as the underlying system evolves.

This section serves two purposes. We first describe the inbound-planning digital twin and the structural dependencies among its principal quantities. We then specify the application-dependent components required by DCC: the candidate calibration mechanisms, the parametric intervention family, the observation likelihood, and the temporal prior linking calibration parameters across prediction horizons.

5.1 Inbound Planning Digital Twin

Amazon uses an Inbound Simulator, a stochastic digital twin of the supply-chain inbound process, to generate weekly multi-horizon forecasts of purchase orders, inbound arrivals, and inventory positions across a large product catalog. These forecasts support downstream procurement, inventory, labor, and capacity planning decisions. At snapshot week t, the simulator maps the current system state, environmental inputs, and stochastic realizations into forecasts over the subsequent H planning horizons:

$$\widehat{\mathbf{Y}}_{t:t+H-1} = S(\mathbf{X}_t; \boldsymbol{\phi}), \tag{5}$$

where $\mathbf{X}_t$ represents the observed system state and external inputs, $\boldsymbol{\phi}$ denotes the simulator's internal mechanisms and parameters, and $\widehat{\mathbf{Y}}_{t:t+H-1}$ denotes the resulting sequence of simulated states and outputs.

Table 2: Mapping from the general DCC framework to the inbound-planning application.

DCC element	Supply-chain instantiation
Entity i	An Amazon Standard Identification Number (ASIN).
Snapshot t	Weekly initialization of the simulator from the current system state.
Horizon h	Week-ahead index, $h \in \mathscr{H} = \{0, \ldots, H-1\}$.
Outcomes $\mathscr{O}$	Experiment-specific quantities used to measure calibration performance.
Candidates $\mathscr{I}$	Observable simulator mechanisms that domain experts identify as both relevant and operationally modifiable.
Observed anchors $\mathscr{O}^{in}$	Candidate mechanisms with aligned production observations.
Intervention	A shared, horizon-specific log-linear transformation of volume.
Structural propagation	Re-execution of downstream simulator logic after an intervention.
Model selection	Prespecified performance on held-out validation data.

The simulator accounts for stochastic demand, vendor behavior, lead times, logistics conditions, and inventory evolution.

Throughout, we use EmuIf Lu et al. (2025), a lightweight emulator of the production inbound simulator. EmuIf replaces selected computationally expensive mechanisms with learned approximations while retaining the surrounding discrete-event logic and the dependency structure among modules, and it reproduces the principal quantities of the inbound process at much lower computational cost. All calibration experiments in Section 6 operate on the emulator.

Table 2 maps the general framework to this application. Throughout this section, *dynamic calibration* refers to variation across look-ahead horizons within a snapshot; reuse of a fitted calibration at later snapshot weeks is termed *temporal generalization*.

The principal quantities used in our experiments are summarized in Table 3. We distinguish these application-specific quantities from the general DCC notation introduced in Table 1.

Table 3: Principal quantities in the inbound-planning digital twin.

Quantity	Meaning
ASIN	Amazon Standard Identification Number, the product-level unit of analysis.
TIP	Target inventory for the planning horizon.
SOQ	Submitted Order Quantity.
COQ	Confirmed Order Quantity.
VLT	Vendor Lead Time.
Open PO	Purchase orders already placed but not yet received.
Future PO	Purchase orders expected to be placed in future planning periods.
Total PO	Total outstanding and future purchase-order volume, equal to Open PO plus Future PO.
Inbound	Units received from vendors.
Outbound	Units leaving inventory to satisfy downstream demand.
Onhand	On-hand inventory available in the system.

At a high level, inventory targets and demand conditions influence the quantities submitted for ordering;

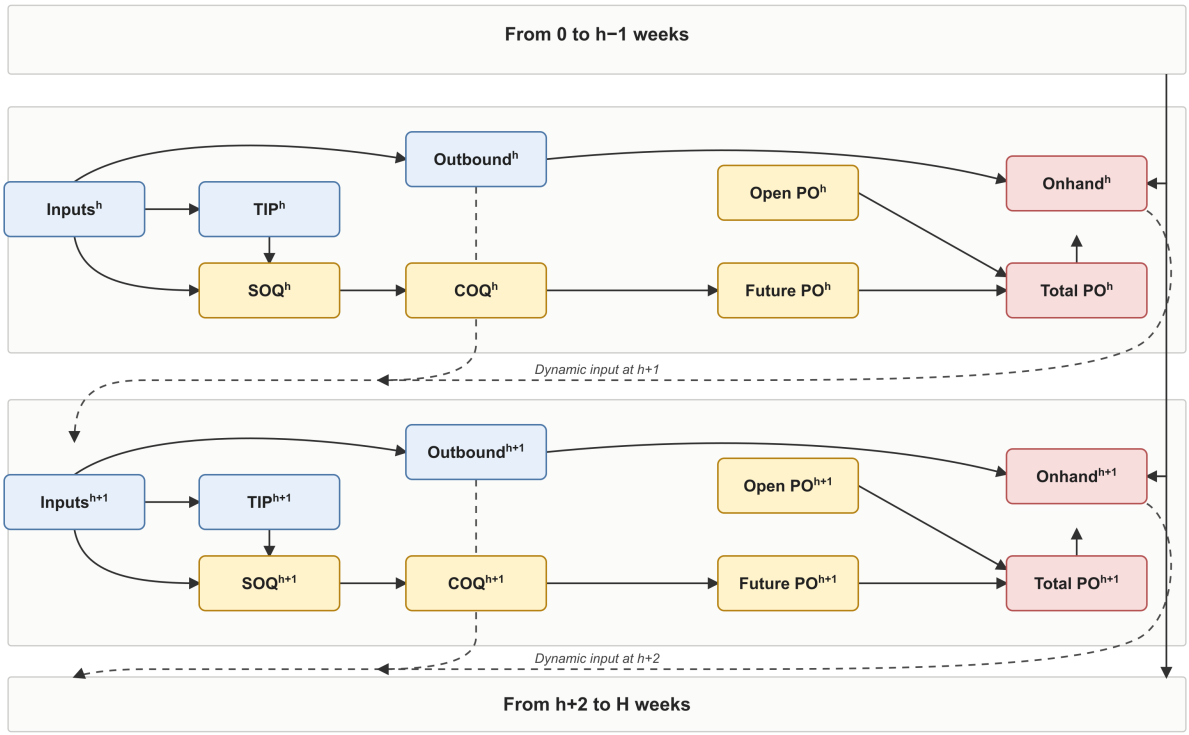


Figure 2: **Dynamic dependency structure of the EmuIf inbound digital twin.** Nodes represent principal quantities in the supply-chain simulator and arrows represent dependencies encoded by its computational logic. Yellow nodes denote observable mechanisms eligible for calibration, red nodes denote target outcomes, and blue nodes denote simulator-internal quantities that are not directly calibrated in the corresponding experiment. Dashed edges represent state propagation across prediction horizons.

vendor behavior determines which orders are confirmed and when they are expected to arrive; open and future purchase orders determine the purchase-order pipeline; and realized inbound and outbound flows drive the evolution of inventory. Consequently, forecast errors introduced at one stage of the simulator can propagate through several downstream quantities and across multiple planning horizons.

The computational graph in Figure 2 was constructed through domain-expert elicitation. Experts familiar with the inbound simulator identified which quantities are read by each mechanism and which downstream quantities are regenerated when that mechanism changes. Temporal feedback was represented by unrolling these dependencies over the finite horizon set $\mathscr{H}$, yielding an acyclic computational graph. The resulting graph is therefore an expert-validated representation of simulator logic, not a causal graph learned from observational data and not, by itself, a causal model of the physical supply chain.

5.2 Structural Dependencies and the Calibration Problem

The inbound simulator is not simply a collection of independently generated forecasts. Its quantities are connected through computational dependencies, accounting identities, and state-transition relationships. For

example, the purchase-order pipeline satisfies

$$\text{TotalPO}_{i,h} = \text{OpenPO}_{i,h} + \text{FuturePO}_{i,h}, \tag{6}$$

while inventory evolves recursively according to relationships of the form

$$\text{Onhand}_{i,h} = \max\{0, \text{Onhand}_{i,h-1} + \text{Inbound}_{i,h} - \text{Outbound}_{i,h}\}. \tag{7}$$

Here i indexes ASINs and h indexes prediction horizons. Equation (6) is an accounting identity used exactly in the simulator. Equation (7) is a schematic statement of the recursive inventory update; the production implementation contains additional operational detail. DCC preserves the implemented transition itself, not merely the abbreviated expression shown here.

These relationships create an important distinction between *output discrepancy* and *mechanism discrepancy*. An observed bias in Total PO, for example, need not originate in the Total PO calculation itself. It may reflect discrepancy in Open PO, submitted or confirmed order quantities, vendor lead times, or other upstream mechanisms. Correcting Total PO directly can improve its marginal accuracy without correcting the mechanism that generated the discrepancy and, if related quantities are calibrated independently, can violate Eq. (6).

Operational implication. The location of calibration matters because the same downstream forecast error can have different upstream causes and therefore different implications for other simulated quantities. A correction introduced upstream can propagate through several downstream outputs, whereas an output-level correction affects only the reported quantity. The supply-chain setting therefore makes concrete the distinction underlying DCC: calibration should identify *where* the simulator departs from observations before determining *how much* that mechanism should be adjusted.

Forecast discrepancy also has direct operational consequences. The simulator's outputs serve as inputs to downstream planning and optimization models, so systematic error can propagate from simulation into operational decisions. At catalog scale, even modest percentage discrepancies can correspond to large absolute volume differences. Moreover, because inventory is a state variable, errors can accumulate across horizons rather than remaining confined to the week in which they originate. These features make structural consistency, scalability, and temporal stability particularly important requirements for production calibration.

5.3 DCC Instantiation for Supply-Chain Volumes

The general DCC framework requires application-specific choices for (i) the mechanisms that may be calibrated, (ii) the parametric intervention family $g_{\boldsymbol{\theta}}$, and (iii) the observation model used to measure agreement with real-world data. For the inbound simulator, these choices are designed for non-negative, heavy-tailed volume quantities with substantial cross-sectional heterogeneity across products and potentially different bias patterns across prediction horizons.

Candidate intervention variables are identified during the same domain-expert elicitation used to construct the graph. A variable is eligible only if (i) its value can be modified before its descendants are evaluated, (ii)

an aligned production observation is available when the variable is used as a direct calibration anchor, and (iii) the intervention is meaningful under the intended deployment workflow. Latent internal states and quantities that are observable but operationally immutable are excluded. The remaining candidates form $\mathscr{I}$; their observed subset forms $\mathscr{O}^{in}$. For each experiment, the final calibration set $\mathscr{C}$ is selected from the parsimonious candidates generated by Section 4.1, which prevents an ancestor and its descendant from entering the same candidate set. Section 6 reports the outcome, candidate, and selected calibration sets for every experiment.

Parametric calibration function

For a calibrated volume variable $X \in \mathscr{C}$, we use a horizon-specific log-linear soft intervention:

$$\log\left(X_{i,h}^{\boldsymbol{\theta}} + 1\right) = b_{X,h} + w_{X,h} \log(X_{i,h} + 1), \qquad w_{X,h} > 0. \tag{8}$$

Here $b_{X,h} \in \mathbb{R}$ is an additive shift in log-space and $w_{X,h} > 0$ controls the response to the original simulated volume. Equivalently,

$$X_{i,h}^{\boldsymbol{\theta}} = \exp[b_{X,h} + w_{X,h} \log(X_{i,h} + 1)] - 1. \tag{9}$$

The parameterization contains several useful special cases. The identity transformation is recovered at

$$b_{X,h} = 0, \qquad w_{X,h} = 1.$$

Setting $w_{X,h} = 1$ produces an approximately multiplicative correction, with $e^{b_{X,h}}$ controlling its magnitude in the transformed scale. Values $w_{X,h} > 1$ amplify cross-sectional differences in simulated volume, whereas $0 < w_{X,h} < 1$ compress them. Thus, the transformation can correct both aggregate level bias and systematic differences in the dispersion of simulated volumes while using only two parameters per variable and horizon.

Importantly, Eq. (8) is applied only at variables in $\mathscr{C}$. DCC does not independently transform their descendants. Instead, calibrated values are inserted back into the simulator and downstream quantities are recomputed through the original structural equations. For example, if Open PO is calibrated while Total PO is not, Total PO is regenerated through Eq. (6); it is not assigned an independent correction factor.

Observation likelihood

Supply-chain volume distributions are non-negative, highly heterogeneous across products, and frequently contain excess zeros. We therefore model observed quantities in log-space using a zero-inflated truncated exponential-power distribution. The model is applied to every observed quantity $Z \in \mathscr{O} \cup \mathscr{O}^{in}$, including both target outcomes and any directly observed calibration anchors. Let $s_{Z,i,h}^{*} = \log(Z_{i,h}^{*} + 1)$ and $m_{Z,i,h}^{\boldsymbol{\theta}} = \log(Z_{i,h}^{\boldsymbol{\theta}} + 1)$. The corresponding mixture likelihood is

$$p_Z(s_{Z,i,h}^{*} \mid m_{Z,i,h}^{\boldsymbol{\theta}}) = \begin{cases} \pi_Z, & s_{Z,i,h}^{*} = 0, \\ (1-\pi_Z) f_{\mathrm{TEP}}\left(s_{Z,i,h}^{*} \mid m_{Z,i,h}^{\boldsymbol{\theta}}, \tau_{Z,h}, \nu_Z\right), & s_{Z,i,h}^{*} > 0, \end{cases} \tag{10}$$

where π_Z is the probability of an observed zero, $\tau_{Z,h} > 0$ is a horizon-specific scale, and $\nu_Z > 0$ controls tail shape. The positive-part density is the exponential-power density truncated to the support of the transformed volume:

$$f_{\mathrm{TEP}}(s \mid m, \tau, \nu) = \frac{f_{\mathrm{EP}}(s \mid m, \tau, \nu)}{1 - F_{\mathrm{EP}}(0 \mid m, \tau, \nu)} \mathbf{1}\{s > 0\}. \tag{11}$$

Here F_{EP} is the cumulative distribution function associated with f_{EP}.

The exponential-power family generalizes the Gaussian by allowing the tail shape to adapt to the empirical distribution:

$$f_{\mathrm{EP}}(s \mid m, \tau, \nu) = \frac{\nu}{2\tau\Gamma(1/\nu)} \exp\left[- \left(\frac{|s-m|}{\tau} \right)^{\nu} \right]. \tag{12}$$

The zero-inflated component accounts separately for observed zeros, while the continuous component measures discrepancy among positive volumes. We allow $\tau_{Z,h}$ to vary by horizon while holding π_Z and ν_Z constant across horizons for parsimony. Distributional hyperparameters are initialized using empirical estimates from the calibration data.

Combining the observation model with the state-space prior below gives the application-specific version of Equation (2):

$$\widehat{\boldsymbol{\theta}}_{\mathscr{C}} = \arg\min_{\boldsymbol{\theta}} \left\{ - \sum_{h \in \mathscr{H}} \sum_{i=1}^{N} \sum_{Z \in \mathscr{O} \cup \mathscr{O}^{in}} \log p_Z \left(s^*_{Z,i,h} \mid m^{\boldsymbol{\theta}}_{Z,i,h} \right) - \log p(\boldsymbol{\theta}) \right\}. \tag{13}$$

Algorithm 1, line 5, solves Equation (13) for each candidate set.

Dynamic state-space prior

The sim-to-real discrepancy need not be constant across the prediction horizon. For example, errors in vendor lead-time or order-generation mechanisms may accumulate as forecasts extend further from the current snapshot. Estimating each horizon independently, however, ignores the strong temporal relationship among neighboring forecast horizons and increases estimation variance.

We therefore connect the horizon-specific calibration coefficients through a Gaussian random-walk state-space prior. Because $w_{X,h} > 0$, its logarithm is modeled as

$$\log w_{X,h} \mid \log w_{X,h-1} \sim \mathscr{N}\left(\log w_{X,h-1}, \sigma^2_{w,X} \right), \tag{14}$$

with

$$\log w_{X,0} \sim \mathscr{N}(0, \sigma^2_{w,X,0}), \tag{15}$$

which centers the initial slope at the identity value $w_{X,0} = 1$.

Because $b_{X,h}$ may be either positive or negative, we model it directly:

$$b_{X,h} \mid b_{X,h-1} \sim \mathcal{N}\left(b_{X,h-1}, \sigma_{b,X}^2\right), \tag{16}$$

with

$$b_{X,0} \sim \mathcal{N}(0, \sigma_{b,X,0}^2), \tag{17}$$

centering the initial shift at the identity value $b_{X,0} = 0$.

The innovation variances $\sigma_{w,X}^2$ and $\sigma_{b,X}^2$ determine how rapidly calibration is allowed to change across prediction horizons. Smaller values encourage nearly constant corrections, while larger values allow stronger horizon-specific adaptation. In this sense, the state-space prior provides partial pooling across horizons: each horizon retains its own calibration parameters, but neighboring horizons borrow information from one another.

Remark 5 (Dynamic calibration versus repeated static calibration). The horizon-dependent parameterization is intended to capture how *forecast discrepancy evolves with look-ahead horizon*, not merely to fit seven unrelated calibration models. The state-space structure occupies a middle ground between a single static correction, which cannot represent horizon-dependent bias, and independent horizon-by-horizon calibration, which can overfit sampling variation. This distinction is especially important in dynamic simulators because upstream discrepancies may accumulate as the simulated state propagates forward.

5.4 Deployment and Evaluation Protocol

The application places three requirements on a useful calibration procedure. First, *structural consistency* requires calibrated outputs to remain compatible with the accounting and state-transition relationships encoded by the simulator. Second, *scalability* requires calibration to operate over large populations of products without introducing a separate correction parameter for every product. Third, *temporal generalization* requires a calibration learned from one system snapshot to remain useful as the physical system evolves.

DCC addresses the first requirement by intervening within the simulator and regenerating descendants through non-intervened structural equations. The shared calibration functions in Eq. (8) address the second: for $|\mathscr{C}|$ calibrated mechanisms over H horizons, the number of calibration coefficients is $O(H|\mathscr{C}|)$ and does not grow with the number of ASINs. The state-space formulation addresses variation across prediction horizons, while generalization across *snapshot time* is an empirical deployment question: a calibration estimated at snapshot t is applied without retraining to later snapshots $t+1, t+2, \ldots$.

These requirements motivate the three levels of evaluation in the next section. The fully synthetic benchmark isolates structural consistency under known ground truth; the semi-synthetic experiments test whether corrections propagate selectively through a nonlinear simulator according to its dependency structure; and the industrial experiment evaluates accuracy, scalability, and out-of-snapshot generalization in the production-scale inbound digital twin.

6 Experiments

We evaluate DCC through three progressively realistic experimental settings, each designed to answer a different methodological question. First, a *fully synthetic benchmark* with known structural equations asks whether DCC can recover accuracy while preserving relationships that independent output calibration violates. Second, *semi-synthetic experiments* replace selected mechanisms in a realistic nonlinear simulator and ask whether calibration effects propagate selectively according to the simulator's dependency structure. Third, *industrial experiments* on Amazon's inbound digital twin evaluate intervention selection, calibration performance, multi-outcome trade-offs, and generalization to future system snapshots.

The progression is deliberate. The synthetic experiments provide control and known ground truth; the semi-synthetic experiments retain realistic nonlinear system behavior while allowing the source of discrepancy to be controlled; and the industrial experiments sacrifice knowledge of the true discrepancy mechanism in exchange for production scale and real sim-to-real observations. Together, the three settings separate evidence about *how DCC works* from evidence about *whether it is useful in practice*.

The experiments are organized around four research questions:

RQ1. Accuracy and structural consistency. Can DCC reduce predictive error while preserving structural identities retained by the calibrated simulator?

RQ2. Structural propagation. When DCC intervenes on selected mechanisms in a nonlinear simulator, do the resulting changes propagate to downstream quantities according to the simulator's dependency structure?

RQ3. Intervention-set selection. Among the candidate mechanisms, can held-out validation identify a useful and parsimonious calibration set for the industrial digital twin?

RQ4. Temporal transfer. Does a calibration estimated at one system snapshot retain its target improvements at later snapshots without retraining?

The experiments support affirmative answers to all four questions within the scopes stated in each subsection; a closing remark delineates what the evidence does not establish.

6.1 Fully Synthetic Benchmark

We begin with a setting in which the true structural equations and the source of simulator bias are known. This experiment isolates the central structural claim of DCC: correcting selected upstream mechanisms and propagating those corrections through the SEM can achieve predictive accuracy comparable to output-wise calibration while retaining the non-intervened structural relationships.

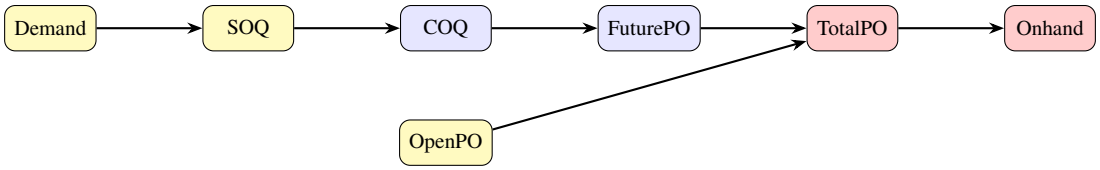


Figure 3: Synthetic benchmark DAG. Yellow nodes indicate variables at which simulator bias is introduced and calibration is applied; blue nodes are regenerated through the structural equations; red nodes are target outcomes.

Data-generating process. The true-world structural equations are

$$\begin{aligned}
\text{SOQ}_{i,h} &= \text{Demand}_{i,h} + 50 + \varepsilon^{\text{SOQ}}, & \text{Demand}_{i,h} &\sim \text{LogNormal}(4.5, 0.8),\\
\text{COQ}_{i,h} &= 0.7\,\text{SOQ}_{i,h} + 10 + \varepsilon^{\text{COQ}}, & \varepsilon^{\text{SOQ}} &\sim \mathcal{N}(0, 10^2),\\
\text{FuturePO}_{i,h} &= 0.8\,\text{COQ}_{i,h} + 5 + \varepsilon^{\text{FPO}}, & \varepsilon^{\text{COQ}} &\sim \mathcal{N}(0, 8^2),\\
\text{OpenPO}_{i,h} &\sim \mathcal{N}(80, 30^2), & \varepsilon^{\text{FPO}} &\sim \mathcal{N}(0, 5^2),\\
\text{Outbound}_{i,h} &\sim \mathcal{N}(120, 40^2), & \text{Onhand}_{i,-1} &\sim \mathcal{N}(500, 150^2),\\
\text{TotalPO}_{i,h} &= \text{OpenPO}_{i,h} + \text{FuturePO}_{i,h},\\
\text{Onhand}_{i,h} &= \max(0,\, \text{Onhand}_{i,h-1} + \text{TotalPO}_{i,h} - \text{Outbound}_{i,h})\,.
\end{aligned}$$

Here $\text{Onhand}_{i,-1}$ is the realized inventory carried into the first prediction horizon, drawn once per entity and held fixed across horizons; the recursion is therefore applied at $h = 0$ as well. Each mechanism draws its own independent noise, and all quantities, including this initial inventory, are truncated at zero. The inventory recursion substitutes Total PO for the inbound flow used in the supply-chain instantiation, which keeps the benchmark's purchase-order pipeline and inventory state coupled through a single quantity.

The simulator uses these same structural equations and the same realizations of every exogenous input, including the same realized initial inventory, but contains known horizon-dependent bias in two upstream mechanisms. The SOQ mechanism is distorted affinely: its noise-free component is multiplied by $1.5(1 + 0.1h)$ and shifted by $20(1 + 0.05h)$, so the implied bias grows from approximately 50% at $h = 0$ to 80% at $h = 6$. Open PO is multiplied by $1.25(1 + 0.1h)$, corresponding to 25%–40% bias. Because every remaining mechanism and every exogenous draw is shared between the two systems, the sim-to-real discrepancy originates only in SOQ and Open PO. Conditions C1 and C2 of Section 3.5 therefore hold by construction with $\mathscr{C} = \{\text{SOQ}, \text{OpenPO}\}$, so Theorem 1(i) guarantees that an exact calibration exists. The benchmark accordingly asks a finite-sample question: whether the estimation procedure of Section 4 recovers such a calibration from data, not whether one exists.

We generate $N = 1{,}000$ instances over $H = 7$ prediction horizons, use an 80/20 train–test split, and repeat the entire experiment over five independent replications.

Methods compared. To separate the effect of *where* calibration is applied from the expressiveness of the calibration map, all methods use simple horizon-specific corrections. We compare:

Table 4: Fully synthetic benchmark: percentage change in MAE relative to the uncalibrated simulator, averaged across seven prediction horizons and reported as mean ± standard deviation over five independent replications. Negative values indicate improvement; bold indicates the best method for each quantity. The Emu MAE column is the mean uncalibrated error across replications.

Metric	Emu MAE	TS Ratio	Indep. Output	DCC
SOQ	188.8	$-94.5\% \pm 0.2$	$\mathbf{-95.0\%} \pm 0.2$	$-90.1\% \pm 0.4$
COQ	132.3	$-92.7\% \pm 0.2$	$\mathbf{-92.8\%} \pm 0.2$	$-89.1\% \pm 0.3$
Open PO	50.2	$\mathbf{-95.0\%} \pm 0.1$	$-94.7\% \pm 0.2$	$-82.8\% \pm 0.8$
Future PO	105.8	$\mathbf{-91.6\%} \pm 0.2$	$-91.5\% \pm 0.2$	$-88.6\% \pm 0.3$
Total PO	156.0	$-93.3\% \pm 0.2$	$\mathbf{-93.6\%} \pm 0.2$	$-92.8\% \pm 0.2$
Onhand	524.7	$-89.5\% \pm 0.1$	$-89.9\% \pm 0.2$	$\mathbf{-95.7\%} \pm 0.1$
Per-instance structural violations		99.9%	99.8%	**0.0%**

- **Time-series ratio correction.** For each metric and horizon, we estimate

$$r_h = \frac{\text{mean}(\text{real}_h)}{\text{mean}(\text{sim}_h)}$$

  on the training instances and apply $\widehat{y}_{i,h} = r_h \, \text{sim}_{i,h}$ to the test instances.
- **Independent output calibration.** A separate log-linear transformation (Eq. 8) is estimated for each reported metric and horizon, without using the simulator's structural dependencies.
- **DCC.** DCC calibrates $\mathscr{C} = \{\text{SOQ}, \text{OpenPO}\}$ and regenerates COQ, Future PO, Total PO, and Onhand through the original structural equations, initialized from the same realized inventory used by the simulator. Regeneration evaluates those equations at their conditional means rather than redrawing the structural noise. Its horizon-specific parameters are linked through the state-space regularization in Section 5.3.

Table 4 exposes the distinction between predictive accuracy and structural consistency. All three calibration approaches substantially reduce marginal error. For Total PO, for example, the three methods achieve MAE reductions within one percentage point of each other, 92.8% to 93.6%. Yet the time-series ratio and independent-output approaches violate $\text{TotalPO} = \text{OpenPO} + \text{FuturePO}$ for essentially every instance because the related quantities are corrected separately. DCC obtains comparable accuracy while producing zero violations of this retained structural equation.

The difference is therefore not that DCC possesses a more expressive calibration function. Indeed, the unconstrained methods are slightly more accurate for the individually generated intermediate quantities, where a free-form per-metric correction can absorb the local bias directly. Rather, DCC restricts the correction to a structurally coherent subset of the forecast space. The experiment illustrates the accuracy–consistency tension introduced in Section 1: unconstrained calibration can optimize each marginal quantity separately, whereas DCC requires improvements to be achievable through the simulator's retained mechanisms.

Remark 6 (Structural consistency is architectural). The zero-violation result follows from the location of calibration rather than from an explicit penalty on constraint violations. Once SOQ and Open PO are modified, Total PO and Onhand are regenerated through their structural equations. Thus, for relationships retained by

the calibrated SEM, consistency is a consequence of the model architecture rather than an additional objective that must be traded against predictive accuracy.

The horizon-level results provide a second observation, and it identifies a setting in which the structural route is not merely more consistent but more accurate. Onhand is the only reported quantity generated by a recursive equation, and it is where the three methods differ most. Averaged across replications, DCC reduces Onhand MAE from 80.1 to 11.7 at $h = 0$ and from 1,091.9 to 31.3 at $h = 6$, so its relative improvement grows from -85.3% to -97.1%. Independent output calibration improves from -70.7% to -93.1% over the same range. The gap widens with look-ahead time: DCC's Onhand error is 2.00 times smaller than independent calibration at $h = 0$ and 2.43 times smaller at $h = 6$.

The reason is structural. An output calibrator estimates a separate correction for the inventory level at each horizon and applies it after the state recursion has already carried a biased level forward, so its residual error compounds along the recursion. DCC instead corrects the upstream mechanisms once and lets the retained transition equation propagate the corrected level, so a single set of $O(H|\mathscr{C}|)$ parameters serves every horizon. This is the empirical motivation for the horizon-dependent parameterization introduced in Section 5.3, and more generally for calibrating inside a dynamic simulator rather than at its outputs.

In this controlled benchmark, DCC reduces MAE by 82.8%–95.7% across the reported quantities, including a 92.8% reduction for the targeted Total PO against 93.3% and 93.6% for the two output-wise baselines, while producing zero violations of the retained purchase-order identity that those baselines violate for essentially every test instance. DCC is also the most accurate of the three methods on Onhand, the one reported quantity generated by a recursive equation, by a margin that widens with the prediction horizon. Predictive improvement therefore does not require sacrificing the tested structural identity when calibration is performed inside the simulator, and on recursively propagated quantities the structural route is more accurate rather than merely more coherent.

6.2 Semi-Synthetic Experiments

The fully synthetic experiment establishes structural consistency under simple known equations. We next ask a different question: *when the simulator is nonlinear and substantially more realistic, do calibration effects still propagate according to its dependency structure?*

To answer this while retaining knowledge of the source of discrepancy, we construct semi-synthetic "real" observations by perturbing selected mechanisms of the supply-chain simulator. In particular, we modify the SOQ mechanism through a nonlinear transformation of its inputs and outputs and shift Open PO through a linear transformation in log-space. Data generated from the modified simulator are treated as observations from the physical system, while DCC is applied to the original simulator. Because the discrepancy-generating mechanisms are controlled by construction, we can compare the observed pattern of improvement with the propagation pattern implied by the simulator DAG.

We consider two calibration configurations. Example 1 targets Total PO using Open PO and Future PO as calibration variables, while Example 2 jointly targets Total PO and Onhand using Open PO and SOQ. Unlike the fully synthetic experiment, the purpose here is not to compare DCC with another calibration function; it is to examine *where changes appear* after an intervention is propagated through a nonlinear simulator.

Table 5: Semi-synthetic Example 1: $\mathscr{O} = \{\text{Total PO}\}$, $\mathscr{C} = \{\text{Open PO, Future PO}\}$, and $\mathscr{O}^{in} = \{\text{Open PO}\}$. Results are averaged across the seven-week prediction horizon.

Avg Week	MAE			AEoA			APB		
	emu	cal	pct%	emu	cal	pct%	emu	cal	bps
SOQ	416	383	−7.8%	2,353,076	2,155,989	−8.4%	764.9%	604.4%	−16,049
COQ	243	194	−20.3%	1,375,407	1,085,757	−21.1%	767.2%	493.9%	−27,335
Future PO	203	269	32.2%	1,148,772	1,519,543	32.3%	691.1%	862.3%	17,127
Open PO	5,551	1,167	−79.0%	31,410,311	6,573,632	−79.1%	99.1%	17.6%	−8,149
Total PO	5,638	1,375	−76.0%	30,261,540	6,640,276	−78.1%	92.7%	20.1%	−7,253
Onhand	22,534	7,274	−67.7%	127,001,436	20,767,184	−83.7%	85.0%	201.4%	11,640

Example 1 shows strong correction at Open PO (−79.0% MAE) and at the target Total PO (−76.0%), with the change continuing downstream to Onhand (−67.7%). Importantly, improvements need not occur uniformly across all variables: Future PO worsens by 32.2%, illustrating that optimizing a structurally coupled target can redistribute discrepancy among its components. Reporting only the target metric would therefore provide an incomplete view of structural calibration.

Table 6: Semi-synthetic Example 2: $\mathscr{O} = \{\text{Total PO, Onhand}\}$, $\mathscr{O}^{in} = \{\text{Open PO}\}$, and $\mathscr{C} = \{\text{Open PO, SOQ}\}$. Results are averaged across the seven-week prediction horizon.

Avg Week	MAE			AEoA			APB		
	emu	cal	pct%	emu	cal	pct%	emu	cal	bps
SOQ	416	67	−84.0%	2,353,076	360,114	−84.7%	764.9%	94.4%	−67,053
COQ	243	40	−83.7%	1,375,407	211,890	−84.6%	767.2%	91.5%	−67,574
Future PO	203	244	20.1%	1,148,772	1,379,659	20.1%	691.1%	860.3%	16,928
Open PO	5,552	3,643	−34.4%	31,410,311	20,606,607	−34.4%	99.1%	74.0%	−2,514
Total PO	5,638	3,756	−33.4%	30,261,540	19,226,949	−36.5%	92.7%	66.9%	−2,575
Onhand	22,534	13,545	−39.9%	127,001,436	72,840,536	−42.6%	85.0%	183.9%	9,885

Example 2 makes the propagation pattern particularly transparent. Calibrating SOQ reduces its MAE by 84.0%, and its downstream quantity COQ improves by 83.7% without being independently calibrated. The target quantities Total PO and Onhand subsequently improve by 33.4% and 39.9%, respectively. Future PO, by contrast, does not share the same improvement and increases by 20.1%. The result illustrates that calibration effects are mediated by the simulator structure rather than imposed independently on every reported output.

DCC does not imply that calibrating one mechanism must improve every quantity in the system. Structural propagation determines *which quantities can respond* to an intervention; the calibration objective determines the direction and magnitude of that response. Consequently, deterioration in a non-targeted quantity is not by itself evidence of calibration failure. It can instead reflect a trade-off induced by the simulator's structural relationships. This observation becomes especially important in the real-data experiments, where the physical system may contain mechanisms absent from the simulator.

In both semi-synthetic configurations, the response follows the simulator's dependency structure rather than appearing as independent output corrections. In Example 1, improvement at Open PO continues to the

targeted Total PO and to downstream Onhand. In Example 2, calibrating SOQ improves the uncalibrated downstream COQ by 83.7%, followed by improvements in Total PO and Onhand. The deterioration of Future PO in both examples shows that reachability does not imply universal improvement: the graph determines where an intervention can propagate, while the fitted objective determines the direction and magnitude of the response.

6.3 Industrial Digital-Twin Experiments

We finally evaluate DCC when neither the true discrepancy mechanism nor a synthetic ground-truth SEM is available. We use the EmuIf emulator of Amazon's inbound supply chain introduced in Section 5.1. The system and its principal quantities were described in Section 5; here we focus on the experimental design and calibration results.

The primary calibration snapshot is 2025-03-30. Each simulator run produces a seven-week prediction horizon for $N = 5{,}520$ high-volume non-sortable ASINs. The unit of analysis is the ASIN: each ASIN contributes one simulated and one observed value per reported quantity at each prediction horizon, so all product-level metrics are averages over ASINs at a fixed horizon. The population is the set of ASINs that pass the high-volume non-sortable screen and have aligned production observations at every one of the seven horizons, which makes the panel complete and avoids averaging over a varying denominator. ASINs are split 80/20 for parameter estimation and intervention-set validation, respectively. The dynamic simulator graph spans order placing (TIP and SOQ), procurement (COQ), the purchase-order pipeline (Open PO, Future PO, and Total PO), and inventory evolution (Onhand).

Evaluation metrics. For predictions $\widehat{y}_i$ and observed values y_i across N ASINs at a given prediction horizon, we report three complementary measures: mean absolute error (MAE), absolute error on aggregate (AEoA), and aggregate percentage bias (APB):

$$\mathrm{MAE}(\mathbf{y},\widehat{\mathbf{y}}) = \frac{1}{N}\sum_{i=1}^{N}|\widehat{y}_i - y_i|, \tag{18}$$

$$\mathrm{AEoA}(\mathbf{y},\widehat{\mathbf{y}}) = \left|\sum_i \widehat{y}_i - \sum_i y_i\right|, \tag{19}$$

$$\mathrm{APB}(\mathbf{y},\widehat{\mathbf{y}}) = \frac{\sum_i \widehat{y}_i}{\sum_i y_i} - 1. \tag{20}$$

MAE measures average product-level discrepancy, AEoA measures the absolute error in total system volume, and APB measures aggregate percentage bias at that horizon, a positive value indicating over-prediction.

Each measure is computed separately for every prediction horizon, and the summary tables report the mean of its absolute values across the seven horizons. For MAE and AEoA this is simply the mean, since both are non-negative by construction. For APB the absolute value is taken first, so the APB columns should be read as a mean absolute percentage bias rather than a net bias: a quantity whose bias changes sign across horizons is summarized by the average magnitude of that bias, not by its partial cancellation. Signed horizon-level discrepancies are reported separately in Table 8. The **pct%** columns report relative change

from the uncalibrated emulation to DCC, so negative values indicate lower error, and the **bps** column reports the change in mean absolute percentage bias in basis points.

Table 7: Uncalibrated emulation-to-real percentage error and absolute volume difference at the 2025-03-30 snapshot, averaged across seven prediction horizons.

	SOQ	COQ	Open PO	Future PO	Total PO	Onhand
pct (abs avg)	57.78%	33.74%	24.50%	58.78%	28.68%	46.11%
vol (abs avg)	939,968	399,255	114,512	394,325	366,622	3,259,970

Table 7 shows substantial baseline discrepancy across the system, but the magnitudes differ considerably by mechanism. In particular, Future PO has substantially larger aggregate percentage discrepancy than Open PO. This heterogeneity is important: a single global correction factor cannot generally address different bias patterns while retaining the accounting relationship between Open PO, Future PO, and Total PO.

For completeness, Table 8 reports the underlying weekly aggregate volume gaps. Bias varies both across quantities and across prediction horizons, providing empirical motivation for the horizon-dependent calibration introduced in Section 5.3.

Table 8: Uncalibrated weekly aggregate volume difference (simulation minus observed value) at the 2025-03-30 system snapshot. The seven rows are the prediction horizons of this single snapshot, not separate snapshots; see Table 12 for the out-of-snapshot comparison.

Target week	SOQ	COQ	Open PO	Future PO	Total PO	Onhand	Outbound
2025-03-30 (h=0)	690,153	267,029	−402,726	217,025	−185,702	−3,707,728	100,797
2025-04-06 (h=1)	671,870	159,847	−169,220	387,837	218,617	−3,601,184	139,126
2025-04-13 (h=2)	1,320,424	574,306	−43,344	434,103	390,759	−3,469,991	226,409
2025-04-20 (h=3)	1,299,110	638,829	−119,162	598,337	479,175	−3,169,913	208,218
2025-04-27 (h=4)	673,166	364,221	36,510	618,003	654,512	−2,576,710	110,429
2025-05-04 (h=5)	474,782	254,116	22,633	334,601	357,233	−2,360,753	172,025
2025-05-11 (h=6)	1,111,374	270,930	−7,992	421,573	413,581	−2,203,708	1,453,320

6.3.1 Selecting the Calibration Set

Before estimating calibration parameters, we determine where calibration should enter the simulator. Following the structural screening procedure in Section 4.1, we construct candidate calibration sets from the manipulable variables

$$\{\text{AvgVLT}, \text{SOQ}, \text{COQ}, \text{OpenPO}, \text{FuturePO}\}.$$

Candidate sets containing selected ancestor–descendant combinations are excluded by the antichain pre-filter, and the remaining configurations are evaluated on the validation split. Table 9 reports the results for the single-target Total PO configuration.

The pair $\mathscr{C} = \{\text{OpenPO}, \text{FuturePO}\}$ achieves the lowest validation nLLK and MAE and is therefore selected for the single-target experiment. Notably, increasing the number of calibrated mechanisms does not

Table 9: Candidate calibration sets for the Total PO experiment evaluated on the validation split. Bold indicates the selected configuration.

	Calibration set $\mathscr{C}$	nLLK	MAE	AEoA
0	**Open PO, Future PO**	**11,266**	**1,046**	**851,593**
1	Open PO, COQ	12,313	1,598	1,049,693
2	Open PO, SOQ	12,749	1,601	1,054,968
3	Open PO	21,474	2,472	2,120,680
4	Open PO, AvgVLT	11,667	2,543	2,240,944
5	Open PO, COQ, AvgVLT	15,017,732	1,570	989,828
6	Open PO, AvgVLT, SOQ	32,622,609,302	1,543	952,308

monotonically improve performance. The three-variable configurations exhibit extremely poor likelihood values despite reasonable MAE, indicating that additional intervention flexibility can create poorly conditioned or over-parameterized calibration configurations.

Remark 7 (More intervention points are not necessarily better). The purpose of intervention-set selection is not to expose every manipulable mechanism to calibration. Each additional intervention introduces another way for the modified simulator to explain the same downstream discrepancy. A parsimonious calibration set therefore serves both computational and statistical purposes: it reduces parameter dimension while limiting competing explanations of the observed sim-to-real gap.

Within the candidate sets evaluated for the single-target experiment, held-out validation selects $\mathscr{C} = \{\text{OpenPO}, \text{FuturePO}\}$ because it achieves both the lowest nLLK and the lowest MAE, and larger candidate sets do not improve these criteria. The evidence therefore supports validation-based selection of a parsimonious set among the considered candidates; it does not establish that this set is the unique physical source of the discrepancy.

6.3.2 Example 1: Calibrating a Single Target

We first target Total PO:

$$\mathscr{O} = \{\text{TotalPO}\}, \qquad \mathscr{O}^{in} = \{\text{OpenPO}\}, \qquad \mathscr{C} = \{\text{OpenPO}, \text{FuturePO}\}.$$

Because Total PO is structurally generated from Open PO and Future PO, DCC modifies its upstream components and regenerates Total PO rather than calibrating it independently.

Table 10: Industrial Example 1: $\mathscr{O} = \{\text{Total PO}\}$, $\mathscr{O}^{in} = \{\text{Open PO}\}$, and $\mathscr{C} = \{\text{Open PO, Future PO}\}$. Results are averaged across the seven-week prediction horizon.

Avg Week	MAE			AEoA			APB		
	emu	cal	pct%	emu	cal	pct%	emu	cal	bps
SOQ	326.8	395.8	21.1%	923,365	1,443,874	56.4%	55.1%	82.0%	2,684
COQ	202.5	274.1	35.4%	363,968	880,666	142.0%	30.9%	70.6%	3,964
Future PO	165.4	104.8	−36.6%	586,358	234,596	−60.0%	102.0%	40.3%	−6,161
Open PO	43.1	42.9	−0.4%	90,591	135,296	49.3%	22.7%	45.6%	2,289
Total PO	189.4	137.2	−27.6%	526,471	365,768	−30.5%	48.6%	33.2%	−1,541
Onhand	700.0	954.4	36.3%	3,031,847	5,288,980	74.4%	42.6%	75.1%	3,249

Table 10 shows a substantial improvement in the targeted purchase order quantity: Total PO MAE decreases by 27.6%, while Future PO MAE decreases by 36.6%. Open PO remains approximately unchanged in product-level MAE (−0.4%). These improvements are achieved while Total PO continues to be generated through its structural relationship with Open PO and Future PO.

The complete system response is equally informative. SOQ and COQ MAE increase by 21.1% and 35.4%, respectively, and Onhand MAE increases by 36.3%. Thus, optimizing one structurally embedded target does not guarantee improvement in every dynamically coupled quantity. In particular, the baseline system under-predicts Onhand, while the calibration that improves purchase-order accuracy can move inventory in the opposite direction. If inventory fidelity is also operationally important, it must be represented in the calibration objective. This motivates the second industrial example.

6.3.3 Example 2: Joint Calibration of Coupled Outcomes

We next expand the outcome set to include both Total PO and Onhand:

$$\mathscr{O} = \{\text{TotalPO}, \text{Onhand}\}, \qquad \mathscr{O}^{in} = \{\text{OpenPO}\},$$

with

$$\mathscr{C} = \{\text{OpenPO}, \text{SOQ}, \text{AvgVLT}\}.$$

This configuration is selected using the same validation procedure as in Section 6.3.1, now evaluating the joint objective for the two outcomes.

Table 11: Industrial Example 2: $\mathscr{O} = \{\text{Total PO, Onhand}\}$, $\mathscr{O}^{in} = \{\text{Open PO}\}$, and $\mathscr{C} = \{\text{Open PO, SOQ, AvgVLT}\}$. Results are averaged across the seven-week prediction horizon.[1]

Avg Week	MAE			AEoA			APB		
	emu	cal	pct%	emu	cal	pct%	emu	cal	bps
SOQ	332.4	501.5	50.9%	968,830	2,635,627	172.0%	57.6%	166.2%	10,861
COQ	206.8	325.5	57.4%	402,107	1,682,991	318.5%	34.0%	144.4%	11,044
Future PO	161.3	167.9	4.1%	550,247	604,083	9.8%	86.6%	92.4%	584
Open PO	43.1	42.4	−1.6%	90,591	103,655	14.4%	22.7%	44.6%	2,194
Total PO	186.3	191.5	2.8%	511,213	527,346	3.2%	47.2%	48.7%	150
Onhand	722.2	678.4	−6.1%	3,278,304	2,746,792	−16.2%	46.2%	38.5%	−765

Including Onhand in the calibration objective changes the operating point. Onhand MAE decreases by 6.1% and its aggregate absolute error decreases by 16.2%, while Total PO MAE increases modestly by 2.8%. The comparison with Example 1 therefore reveals a genuine multi-outcome trade-off: a calibration configuration optimized primarily for purchase-order fidelity differs from one that also values inventory fidelity.

This trade-off is structurally interpretable. Total PO is over-predicted on average in the baseline simulator, whereas Onhand is substantially under-predicted. Adjustments that reduce one discrepancy can therefore move the other in an unfavorable direction through the inventory dynamics. The modest Onhand improvement also indicates a boundary of calibration through the current simulator structure: observed inventory is affected by mechanisms not fully represented in the digital twin, including manually initiated and vendor-initiated orders. We return to this issue in Section 7.3.

Remark 8 (Calibration can reveal model boundaries). A structurally constrained calibration cannot freely move every outcome toward its observed value. When discrepancies in multiple outcomes cannot be simultaneously reconciled through the mechanisms represented in the simulator, the resulting trade-off can indicate missing mechanisms, misspecified dependencies, or competing calibration objectives. DCC can therefore serve not only as a correction layer but also as a diagnostic of where the simulator's existing structure is insufficient to explain the observed system.

6.3.4 Generalization to Future System Snapshots

The preceding experiments evaluate calibration at the snapshot from which its parameters are estimated. For a digital twin used repeatedly in production, a second question is whether those parameters capture a persistent discrepancy or merely fit snapshot-specific noise.

We therefore freeze the Example 1 calibration parameters estimated at the 2025-03-30 snapshot and apply them, without retraining, to five later evaluated snapshots through 2025-05-11. This experiment is

[1]Both industrial examples are evaluated on the held-out validation partition, and Examples 1 and 2 were executed as separate calibration jobs, each drawing its own 80/20 estimation and validation split of the 5,520 ASINs. The uncalibrated columns of Tables 10 and 11 are therefore computed on slightly different subsets of products and differ accordingly, for example SOQ MAE 326.8 against 332.4 and Onhand MAE 700.0 against 722.2, even though both describe the same simulator at the same snapshot. Within each table the **pct%** and **bps** columns compare calibrated against uncalibrated quantities on a common partition, and the cross-example comparison in the text is made between these relative changes rather than between absolute levels.

distinct from the horizon-dependent state-space model in Section 5.3. The latter smooths parameters across prediction horizon h *within* a snapshot; the present experiment evaluates transfer across snapshot time t.

Table 12: Out-of-snapshot generalization: percentage change in MAE after applying the calibration learned on 2025-03-30 without retraining. Each row is a separate simulator run initialized at that week and producing its own seven-week prediction horizon; the reported change is averaged across that horizon. Negative values indicate improvement.[2]

Snapshot	SOQ	COQ	Future PO	Open PO	Total PO	Onhand
2025-03-30 (t)	+21.1	+35.4	**−36.6**	−0.4	**−27.6**	+36.3
2025-04-06 (t+1)	+22.2	+41.0	**−37.6**	−3.1	**−29.2**	+45.8
2025-04-20 (t+3)	+20.8	+32.4	**−34.0**	+2.5	**−25.3**	+48.1
2025-04-27 (t+4)	+21.6	+28.9	**−29.8**	+1.2	**−21.5**	+44.8
2025-05-04 (t+5)	+19.7	+33.3	**−28.6**	+0.0	**−19.9**	+45.2
2025-05-11 (t+6)	+14.2	+23.3	**−30.6**	+3.2	**−20.9**	+44.3

The target behavior is stable across the evaluated snapshots. Future PO MAE improves by 28.6–37.6%, and Total PO MAE improves by 19.9–29.2%, despite the absence of retraining. At the same time, the non-targeted quantities retain the same qualitative trade-off observed at the training snapshot: SOQ, COQ, and Onhand worsen. The persistence of both the benefit and the trade-off is important. It suggests that the learned intervention is capturing a systematic sim-to-real discrepancy rather than simply exploiting idiosyncratic noise in the calibration snapshot.

The experiment also clarifies two distinct notions of dynamics in DCC. Across prediction horizon h, calibration parameters are allowed to evolve smoothly through the state-space prior because forecast bias can change with look-ahead time. Across system snapshots t, by contrast, no adaptation is performed in Table 12: the parameters are deliberately frozen. The observed performance therefore provides evidence of short-term temporal transfer rather than being a consequence of continual recalibration. The evaluation window is nevertheless limited and does not establish robustness to seasonal changes, demand shocks, or structural changes in the underlying system.

Over the five later snapshots evaluated here, and without retraining, the calibration reduces Future PO MAE by 28.6–37.6% and Total PO MAE by 19.9–29.2% at every evaluated future snapshot. The result supports short-term temporal transfer of the target improvements, while the persistent deterioration in SOQ, COQ, and Onhand shows that the associated non-target trade-offs also transfer. Because the window is short, the experiment does not establish long-term robustness under seasonal or structural change.

Remark 9 (What DCC is and is not). Across the three settings, DCC operates as a structure-preserving calibration layer: it corrects selected mechanisms and lets the retained simulator equations regenerate their descendants. It does not guarantee universal improvement. Structural propagation does not imply that every coupled quantity improves, and preserving the simulator's retained equations does not make an incomplete simulator structurally complete. DCC restricts calibration to corrections that can be expressed through the

[2]The snapshot dates listed here index simulator *initializations*, whereas the dates in Table 8 index the seven prediction *horizons* of the single 2025-03-30 snapshot; the two sequences coincide numerically but refer to different quantities. No snapshot initialized at t+2 (2025-04-13) is included, so five later snapshots are reported rather than six.

modeled mechanisms; this restriction is the source of its consistency advantage, but it also makes missing mechanisms and multi-outcome tensions visible. Section 7.3 discusses these limitations and their implications for extending DCC beyond the present application.

7 Conclusions and Future Directions

7.1 Main Findings

The central finding is that simulator calibration should be evaluated not only by marginal predictive error but also by whether the resulting outputs remain compatible with the simulator relationships required for downstream use. DCC addresses these two objectives together by calibrating selected mechanisms and regenerating their descendants through the retained simulator equations.

The evidence can be summarized through the four research questions posed in Section 6. On accuracy and structural consistency (RQ1), the fully synthetic benchmark shows that DCC reduces MAE by at least 82.8% across the reported quantities and by 92.8% for Total PO, against 93.3% and 93.6% for the two output-wise baselines, while producing zero violations of the tested purchase-order identity that those baselines violate for essentially every instance. The benchmark also separates two regimes: output-wise calibration is marginally more accurate on quantities produced by a single mechanism, whereas DCC is the most accurate method on the recursively propagated inventory level, where its error is 2.0 times lower than independent output calibration at the first horizon and 2.4 times lower at the seventh.

On structural propagation (RQ2), the semi-synthetic experiments show that corrections propagate selectively through a nonlinear simulator. For example, calibrating SOQ reduces MAE in the uncalibrated downstream COQ by 83.7%, followed by improvements in Total PO and Onhand. Other quantities deteriorate, confirming that structural reachability does not imply universal improvement.

On intervention-set selection (RQ3), held-out validation favors a parsimonious two-mechanism set for the single-target industrial experiment. The selected calibration reduces Total PO MAE by 27.6%, although several non-target quantities deteriorate. This result supports validation-based selection among the considered candidates but does not identify a unique physical cause of discrepancy.

On temporal transfer (RQ4), the same calibration, applied without retraining, reduces Total PO MAE by 19.9–29.2% at each of five later evaluated snapshots. The persistence of both the target benefit and the non-target trade-offs supports short-term transfer, not robustness to arbitrary seasonal or structural change.

The two settings also differ instructively on the recursively propagated inventory level. In the synthetic benchmark, where the discrepancy is confined to the calibrated mechanisms by construction and conditions C1 and C2 of Section 3.5 therefore hold, structural propagation makes DCC the most accurate method on that quantity. In the industrial twin it does not, and observed inventory is known to be affected by mechanisms absent from the simulator, including manually initiated and vendor-initiated orders. Under Remark 2 this is the expected signature of a violated C1: the discrepancy cannot be expressed through the retained structure, so no parameter value reconciles it.

Together, these results establish the paper's principal empirical claim: DCC can narrow target sim-to-real gaps while preserving the simulator equations through which those targets are regenerated. They also expose

the corresponding boundary: structural coherence constrains which combinations of observed quantities can be matched through the selected mechanisms.

7.2 Implications for Simulation Practice

Output accuracy and structural fidelity are distinct properties. Two calibration procedures can achieve similar marginal accuracy while producing different joint system states. When several simulator outputs jointly feed planning or optimization models, correcting them independently can therefore create forecasts that cannot coexist under the accounting identities, state-transition rules, or other relationships required by downstream users. A structure-aware workflow instead calibrates selected mechanisms, recomputes their descendants, and evaluates both targeted and non-targeted quantities.

The resulting trade-offs are informative but must be interpreted carefully. DCC reveals whether the *selected mechanisms and calibration family* can reconcile the observed quantities while retaining the specified simulator equations. Failure to reconcile them does not by itself prove that a physical mechanism is missing: it may also reflect an incomplete candidate set, an insufficient calibration family, limited observations, or optimization failure. Persistent residual discrepancy should therefore trigger investigation of these alternatives before the simulator structure is revised.

DCC also extends the simulation-fidelity tradition represented by Nelson's work on input-model uncertainty. Input modeling concerns what enters a simulator; DCC concerns how selected internal mechanisms transform those inputs into coupled outputs. The approaches are complementary: accurate inputs, appropriate stochastic models, uncertainty quantification, and structurally coherent calibration address different dimensions of fidelity in an operational digital twin.

7.3 Limitations and Future Research

The present evidence has several empirical limitations. The industrial study examines one supply-chain application, and its same-snapshot results are reported on the validation split used to select the intervention set. The study does not include industrial comparisons with independent output calibration or a static version of DCC, and it reports point estimates without repeated-split or bootstrap uncertainty intervals. Temporal transfer is evaluated over only five later snapshots. The semi-synthetic experiments cover a limited class of controlled discrepancies, and practical scalability is demonstrated through parameter dimension and catalog-scale application rather than a systematic runtime benchmark. These features limit claims about comparative superiority, statistical uncertainty, long-term stability, and performance in other domains.

The method also assumes that the relevant computational dependency structure is known. In this application the graph is elicited from domain experts, but uncertainty or disagreement in that elicitation is not modeled. The graph records dependencies in the simulator and is not, without additional assumptions, an identified causal graph of the physical system. Future work could test, learn, or revise portions of the graph jointly with calibration and study sensitivity to alternative dependency structures.

Intervention-set selection presents a related identification problem. The selected set is parsimonious among the candidates examined, but it is not necessarily the unique or globally minimal explanation of the

observed discrepancy. Different interventions may induce similar downstream distributions, particularly when only a subset of internal quantities is observed. Establishing identifiability conditions and principled comparisons among competing intervention structures is therefore a priority for developing DCC from a predictive calibration procedure into a stronger diagnostic method.

More generally, four kinds of uncertainty remain: uncertainty in the intervention location, the calibration parameters, the elicited simulator graph, and future system evolution. The current MAP implementation provides a computationally practical point estimate but does not quantify these sources jointly. Scalable Bayesian, likelihood-free, or simulation-based inference could provide predictive intervals and posterior support for alternative mechanisms. Such uncertainty quantification is particularly important when the calibrated simulator feeds consequential planning or optimization decisions.

Dynamics across calendar time provide another priority. The present state-space model varies parameters across look-ahead horizon within one snapshot; it does not update them across snapshots. A two-dimensional model over horizon and calendar time could support drift monitoring and help distinguish routine recalibration from evidence that the simulator itself requires revision. Longer evaluations would be needed to assess seasonality, demand shocks, policy changes, and changes in simulator logic.

Finally, DCC is formulated for structured stochastic simulators, but its industrial evidence comes from a single supply-chain setting. Applications in manufacturing, transportation, service systems, and healthcare operations would test its empirical generality. The present paper provides a practical framework for correcting selected simulator mechanisms, propagating those corrections through retained computational relationships, and exposing accuracy–consistency trade-offs that independent output calibration can conceal. Broader statistical guarantees, runtime comparisons, and cross-domain validation remain open.

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